

HW06 — AI-Assisted API Testing Main Report

Student ID: 23127194
Assignment: HW06 — API Testing
Date: 2026-08-22
SUT: EShop (ttbhanh/eshop-sut)
Execution toolchain: Postman + Newman
AI workflow: AI-first with Human-in-the-Loop review gates

1. Assignment Overview

This assignment applies an AI-assisted, human-reviewed API-testing workflow to three APIs selected from three different pools. For every API, the work includes requirement analysis, domain partitioning, state/security/schema analysis, AI testcase generation, human audit, human-added coverage, Postman implementation, real Newman execution, and defect confirmation from execution evidence.

The three assigned APIs are:

API	Pool / FR	Method and endpoint	Main testing focus
API 1	Pool A / FR-05	GET /api/products?search=keyword	Search input domains, injection/parser behavior, response consistency
API 2	Pool B / FR-10	PUT /api/orders/:id/cancel	Order state machine, invalid transitions, authorization/ownership
API 3	Pool C / FR-16	POST /api/admin/import-products	Import validation, atomicity, admin authorization, persistence integrity

The final suite contains **152 testcase records: 136 AI-generated and 16 human-added**. The official Newman executions ran **141 testcase IDs; 11 FR-16 records remained blocked** because the required CSV transport or safe supporting runtime endpoint/error trigger was unavailable.

2. SUT and Test Environment

The SUT was executed locally on <http://localhost:3000>. The homework repository stores the specification, per-API analysis, generated/audited testcases, Postman collections, runtime data, raw Newman reports, screenshots, bug reports, CI/CD evidence, AI audit, and the AI-driven generator artifacts.

Every official Postman/Newman request was required to carry:

X-Student-Id: 23127194

The final official evidence confirms exact header coverage on **353/353 requests**:

API	Requests	X-Student-Id exact match
FR-05	62	62/62

API	Requests	X-Student-Id exact match
FR-10	96	96/96
FR-16	195	195/195
Total	353	353/353

Raw Newman JSON/HTML/CLI/JUnit artifacts are preserved under `postman/newman/`. Compact execution summaries are used for analysis without replacing the raw reports as the source of truth.

3. AI Usage Method and Human Review

The workflow deliberately avoided a one-prompt “generate and run everything” approach. Instead, each API moved through separate stages:

Requirement Extraction

→ Human Gate A

→ Domain Partition

→ Human Gate B

→ State Transition

→ Human Gate C

→ Security Analysis

→ Human Gate D

→ Schema Analysis

→ AI Test Generation

→ Human Audit

→ Human Extension

→ Postman Implementation

→ Runtime Preparation

→ Real Newman Execution

→ Failure Analysis

→ Human Bug Confirmation

AI was used to produce breadth and structure. Human review was responsible for preserving unresolved requirements, correcting non-deterministic tests, adding missing cross-request scenarios, and deciding whether observed runtime failures represented genuine SUT defects.

The detailed chronological record is maintained in `reports/ai-audit-report.md` and summarized again in Section 13 / Appendix A of this report.

4. API Selection

The selected combination satisfies the assignment requirement of exactly one API from each of Pools A, B, and C:

1. FR-05 Product Search — GET `/api/products?search=keyword`

2. FR-10 Cancel Order — PUT `/api/orders/:id/cancel`

3. FR-16 Import Products — POST `/api/admin/import-products`

Selection details and endpoint/security mapping are recorded in `reports/api-selection.md` and `reports/api-inventory.md`.

5. API 1 — FR-05 Product Search

5.1 Requirement and Test Design Analysis

FR-05 was analyzed across query-domain behavior, response semantics, parser/input edge cases, security risks, and state effects caused by related product operations. The API itself has no explicit business state machine, so the state analysis focused on resource existence/mutation and sequences that could affect later search results.

Artifacts:

- PoolA-FR-05-ProductSearch/analysis/requirements.md
- PoolA-FR-05-ProductSearch/analysis/domain-partitions.md
- PoolA-FR-05-ProductSearch/analysis/state-transitions.md
- PoolA-FR-05-ProductSearch/analysis/security.md
- PoolA-FR-05-ProductSearch/analysis/schema.md

5.2 AI Generation and Human Audit

The AI generated **42** FR-05 cases. The initial human audit classified:

VALID	INCOMPLETE	INVALID	Total
32	10	0	42

The ten incomplete cases were preserved, corrected into deterministic forms, and human-reviewed again. Final audit status was **42 VALID / 0 INCOMPLETE / 0 INVALID**.

This is an important audit result: AI's main weakness was not irrelevant test ideas, but insufficiently concrete inputs/oracles such as vague "long input" or injection probes.

5.3 Human Extension

Humans added **5** cases, bringing the FR-05 executable testcase set to **47**. These cases targeted parser edge conditions, cross-request persistence, and unsupported/unauthorized write behavior that the endpoint-scoped AI generation did not cover deeply enough.

The most important human-added cases were:

- HUMAN-FR05-044 — encoded null-byte parser/injection probe.
- HUMAN-FR05-048 — before/after search plus unauthenticated malformed **POST** sequence.

5.4 Postman / Newman Execution

Official result:

Metric	Result
Testcase IDs executed	47
Passed testcase IDs	45
Failed testcase IDs	2

Metric	Result
Requests	62
Assertions	142
Assertions passed	138
Assertions failed	4
Student header	62/62

The two failed testcase IDs were [HUMAN-FR05-044](#) and [HUMAN-FR05-048](#).

5.5 Confirmed Bugs

Human Gate H confirmed **3 defects**:

Bug	Severity	Confirmed behavior	Found by
FR05-BUG-01	High	Search input reaches SQL construction unsafely and a null-byte probe exposes an SQLite parser error	HUMAN-FR05-044
FR05-BUG-02	Critical	POST /api/products accepts a request without JWT/Admin authorization and persists a product	HUMAN-FR05-048
FR05-BUG-03	High	Product creation accepts missing required fields and persists an all-null record	HUMAN-FR05-048

GitHub Issues: [#24](#), [#25](#), [#26](#).

These three defects are counted as **AI-missed bugs** because their confirming scenarios were human-added rather than AI-generated.

6. API 2 — FR-10 Cancel Order

6.1 Requirement and State Analysis

FR-10 is primarily a state-transition API. The contract permits user cancellation only from cancelable order states and requires invalid transitions to be rejected without changing the order state. Security analysis also considered authentication, ownership/IDOR risk, malformed IDs, repeated cancellation, and response characterization when exact error status/schema were not specified.

Artifacts are stored under [PoolB-FR-10-CancelOrder/analysis/](#).

6.2 AI Generation and Human Audit

The AI generated **42** cases. Human audit identified **6 INCOMPLETE** cases that required more reproducible setup or evidence/oracle definitions. After correction and explicit human re-review, all **42 AI cases were implementation-ready**; none was discarded as INVALID.

6.3 Human Extension

Humans added **6** additional cases. The final FR-10 execution set therefore contained **48 testcase IDs**.

The human additions concentrated on risk-based authorization, state-history dependencies, sequencing, and gaps that required more than isolated single-request reasoning.

6.4 Postman / Newman Execution

Official result:

Metric	Result
Testcase IDs executed	48
Passed testcase IDs	45
Failed testcase IDs	3
Requests	96
Assertions	275
Assertions passed	269
Assertions failed	6
Student header	96/96

The failed testcase IDs were:

- AI-FR10-003
- AI-FR10-021
- AI-FR10-040

All three independently demonstrated the same defect.

6.5 Confirmed Bug

FR10-BUG-01 — User can cancel an order in shipping state (High).

Expected: a user cancellation of a shipping order must be semantically rejected and the order must remain shipping.

Actual: the cancel endpoint returned HTTP 200, and follow-up reads showed the order persisted as canceled. The behavior reproduced on three independently isolated shipping fixtures.

GitHub Issue: #27.

This defect was discovered by **AI-generated tests** and therefore is not counted as AI-missed.

7. API 3 — FR-16 Import Products

7.1 Requirement and Contract Analysis

FR-16 required special handling because two input surfaces had to be distinguished:

1. the business CSV-import contract; and
2. the documented JSON products[] API contract.

The analysis avoided inventing exact HTTP status codes or response field names when the specification did not define them. Persistence became the main oracle for validation, authorization, and atomic rollback. The critical requirement was that invalid rows must not produce partial batch persistence.

Artifacts are stored under `PoolC-FR-16-ImportProducts/analysis/`.

7.2 AI Generation and Human Audit

The AI generated **52** cases, exceeding the minimum requirement of 35. Initial human review found seven incomplete cases. After correction/re-review, the final audit state was:

VALID	INCOMPLETE / BLOCKED	INVALID	Total
51	1	0	52

`AI-FR16-046` remained blocked because no safe, deterministic DB-level error trigger could be established. The case was not fabricated or force-executed.

7.3 Human Extension

Humans retained **5** FR-16 cases after reviewing six candidates. The retained tests covered concurrency, stale category references, token lifecycle, stale role/token state, and repeated invalid submissions.

Total FR-16 testcase records: **57**.

7.4 Runtime Readiness and Blocked Cases

Only **46 of 57** FR-16 testcase records were executable in the official environment. **11 were marked NOT EXECUTED — BLOCKED**, including cases that depended on a faithful CSV transport, safe DB-error trigger, logout/revocation endpoint, or role-promotion endpoint.

Keeping these cases blocked preserves evidence integrity: unavailable runtime capabilities were not simulated and no fake execution result was created.

7.5 Postman / Newman Execution

Official result:

Metric	Result
Testcase IDs executed	46
Passed testcase IDs	29
Failed testcase IDs	17
Blocked testcase records	11
Requests	195
Assertions	388
Assertions passed	342
Assertions failed	46
Student header	195/195

The failures clustered around three real defect classes: invalid price acceptance, violation of all-or-nothing rollback, and non-admin access to the admin import operation.

7.6 Confirmed Bugs

Human Gate H confirmed **3 defects**:

Bug	Confirmed behavior
FR16-BUG-01	Import accepts non-positive product prices
FR16-BUG-02	Import violates atomic all-or-nothing rollback
FR16-BUG-03	Non-admin user can import products through the admin endpoint

GitHub Issues: #28, #29, #30.

The primary FR-16 defect classes were already represented by AI-generated tests, so they are not counted as AI-missed bugs. A human repeated-invalid-batch case (**HUM-FR16-006**) also reproduced rollback-related behavior, but this is the same defect class rather than an additional independent bug.

8. Postman Features Used

The final implementation uses Postman/Newman features for real testing rather than checklist-only demonstrations:

- separate collections for FR-05, FR-10, and FR-16;
- public environment files plus git-ignored private runtime environments;
- collection/environment/local variables;
- collection-level pre-request scripts;
- mandatory **X-Student-Id** injection;
- runtime JSON data files;
- test scripts and semantic/persistence assertions;
- helper requests for state verification;
- **pm.sendRequest()** where cross-request/concurrency behavior required it;
- HTML, JSON, CLI, and JUnit Newman reporters.

The shared runner is:

```
./scripts/run-step-k.sh FR05
./scripts/run-step-k.sh FR10
./scripts/run-step-k.sh FR16
```

9. CI/CD Integration

The repository includes **.github/workflows/hw06-api-tests.yml** and a deliberately separate deterministic CI demonstration suite. This separation prevents genuine official Step-K defect failures from being hidden merely to produce a green pipeline sample.

Run A — all passing

- Commit: `06cd7d9b544a3d0ee9d8c5d19e5d00f1af4bba06`
- GitHub Actions run: `32492084678`
- Result: **SUCCESS**
- Requests: 4
- Assertions: 12
- Failed: 0
- Student header: 4/4

Run B — exactly one intentional failure

- Commit: `b564ce2229ada6f4c14a13491eaafd37086b6068`
- GitHub Actions run: `32492204368`
- Result: **FAILURE — intentional CI demonstration**
- Requests: 4
- Assertions: 13
- Failed: exactly 1
- Failing assertion: `CI-DEMO-ONLY | intentional single failure`

The branch was then restored to passing mode in commit `582811c82d7239c91a819d938e4dd36696423467`, with successful run `32492332236`.

Detailed evidence: `cicd/CI-CD-Report.md`.

10. AI-Driven API Test Generator

The P12/P13 generator artifacts are under `agent-skill/`:

- `agent-skill/pseudocode.md`
- `agent-skill/api-test-generator/SKILL.md`
- `agent-skill/api-test-generator/generate.mjs`
- `agent-skill/api-test-generator/README.md`
- `agent-skill/api-test-generator-diagram.mmd`
- `agent-skill/api-test-generator-diagram.png`
- `agent-skill/demo/`

The generator accepts an API specification, endpoint, FR/security context, and output format, then produces an auditable requirement/endpoint/coverage model and candidate tests. It deliberately preserves unknowns instead of inventing unsupported contract details and ends at a **Human Review Required** gate.

Its internal flow is:

```
Specification Input
→ Requirement Parser
→ Endpoint Model
→ Parameter / Constraint Analyzer
→ Domain Partition Generator
→ State Transition Analyzer
→ Security Mapper
```


- Schema Test Generator
- Coverage Analyzer
- Deduplicator
- Human Review Gate
- Final Test Cases

11. Consolidated Test Summary

11.1 Testcase-level summary

The assignment README requirement refers to tests/testcases, so testcase IDs—not Newman assertions—are used for the main PASS/FAIL totals.

Metric	FR-05	FR-10	FR-16	Total
APIs	1	1	1	3
AI-generated testcases	42	42	52	136
Human-added testcases	5	6	5	16
Total testcase records	47	48	57	152
Executed testcase IDs	47	48	46	141
Passed testcase IDs	45	45	29	119
Failed testcase IDs	2	3	17	22
Blocked / not executed	0	0	11	11
Confirmed bugs	3	1	3	7
AI-missed confirmed bugs	3	0	0	3

Check: 119 passed + 22 failed = 141 executed; 141 executed + 11 blocked = 152 total testcase records.

11.2 Newman assertion-level evidence

Newman assertion numbers are preserved separately because one testcase can contain multiple assertions:

Metric	FR-05	FR-10	FR-16	Total
Requests	62	96	195	353
Assertions	142	275	388	805
Passed assertions	138	269	342	749
Failed assertions	4	6	46	56

The assertion totals must not be confused with testcase PASS/FAIL counts.

12. AI Critique

The evidence-based critique is maintained in [reports/AI-Critique.md](#) and is summarized here.

AI was effective at generating broad systematic coverage, but first-pass outputs frequently needed human correction to become executable. FR-05 had ten initially incomplete AI cases, FR-10 had six, and FR-16 had seven. The recurring problem was insufficient reproducibility: vague long strings, abstract injection payloads, or environment-dependent error triggers needed concrete values, fixtures, and branching oracles.

The strongest AI blind spot was compositional and cross-endpoint reasoning. Human-added FR-05 tests combined parser-layer attacks with before/after persistence checks and revealed three confirmed bugs that the AI suite had not identified. Human additions elsewhere covered concurrency, stale references, token lifecycle, role-state consistency, and repeated operations. These gaps reflect the endpoint-scoped generation process: it was strong at local partitions and state/security matrices but weaker at temporal and cross-feature interactions.

At the same time, AI-generated cases did successfully reveal the FR-10 shipping cancellation defect and the main FR-16 validation, rollback, and authorization failures. The practical lesson is that AI is useful for systematic breadth, while humans remain necessary to challenge assumptions, turn vague tests into reproducible experiments, add cross-layer sequences, and decide whether observed failures actually violate the specification.

13. AI Audit

The complete chronological AI audit is preserved in:

[reports/ai-audit-report.md](#)

The audit was compiled without recreating missing historical transcripts or fabricating evidence. Before the P14 compilation entry, it contained **59 recorded interaction occurrences / 56 distinct historical IDs**, covering setup/shared workflow, all three per-API pipelines, consolidation/CI, the generator, and the diagram process. The compilation normalized equivalent historical field labels while preserving numbering anomalies rather than rewriting history.

The audit demonstrates the required pattern:

AI Output

→ Human Evaluation

→ Correction / Decision

→ Final Artifact

Representative chains include:

Audit chain	Human control demonstrated
AI-002	Generalized an initially HW06-specific testing skill instead of accepting hard-coded assumptions
AI-005 → AI-006	Preserved unresolved FR-05 behavior rather than inventing search semantics

Audit chain	Human control demonstrated
AI-029 → AI-030	Corrected six incomplete FR-10 cases and re-approved them explicitly
AI-043 → AI-045	Corrected/re-reviewed FR-16 incomplete cases; retained a blocked DB-error case when no safe trigger existed
AI-054	Separated truthful defect-bearing Step-K suites from the controlled CI pass/one-fail demonstration
AI-055	Fixed generator parsing after validation exposed a false HTTP-status interpretation
AI-056	Human-reviewed the diagram artifact before final rendering

Evidence-integrity constraints applied throughout the audit:

- no fake Newman execution;
- no fabricated screenshot;
- no invented GitHub Actions result;
- no fake human approval;
- no defect claimed without specification + reproduction + execution evidence;
- blocked cases remain blocked rather than being assigned artificial PASS/FAIL results.

For submission, [reports/ai-audit-report.md](#) is the authoritative detailed AI-audit appendix accompanying this main report.

14. Conclusion

HW06 completed the required three-API AI-assisted testing workflow across Pools A, B, and C. The work produced **152 testcase records**, executed **141** real testcase IDs, preserved **11 blocked cases** without fabricating evidence, and confirmed **7 genuine defects**. Three of those defects were found only after human-added FR-05 scenarios, while AI-generated tests successfully exposed the FR-10 state-transition bug and the principal FR-16 validation/transaction/authorization failures.

The results support the central human-in-the-loop design of the assignment: AI improves coverage breadth and analysis speed, but reliable testing still depends on human review of requirements, deterministic setup, cross-request reasoning, evidence integrity, and final bug confirmation.

Appendix A — Key Evidence Index

Area	Primary artifact
API selection	reports/api-selection.md
API inventory	reports/api-inventory.md
Consolidated test cases	HW06-API-Test-Cases.xlsx
FR-05 audit	PoolA-FR-05-ProductSearch/audit/audit-summary.md
FR-05 execution	postman/newman/FR05-execution-summary.md

Area	Primary artifact
FR-10 execution	postman/newman/FR10-execution-summary.md
FR-16 execution	postman/newman/FR16-execution-summary.md
Full Newman evidence	postman/newman/*-official-report.{json,html,xml}
Confirmed bug reports	Pool*-*/bugs/
CI/CD	cicd/CI-CD-Report.md
AI generator	agent-skill/
AI audit	reports/ai-audit-report.md
AI critique	reports/AI-Critique.md

Appendix B — Repository / External Evidence

- Repository: <https://github.com/nhphunng/SoftwareTesting-Homework>
- Confirmed bug issues: [#24](#) through [#30](#)
- CI all-pass run: <https://github.com/nhphunng/SoftwareTesting-Homework/actions/runs/32492084678>
- CI one-fail demonstration: <https://github.com/nhphunng/SoftwareTesting-Homework/actions/runs/32492204368>
- CI restored-pass run: <https://github.com/nhphunng/SoftwareTesting-Homework/actions/runs/32492332236>

Appendix C — Full AI Audit Report

Declaration: I use AI tools for the following tasks.

P14 Compilation Status

- Compiled on: [2026-08-21T22:27:00+07:00](#)
- Historical corpus before this P14 compilation entry: **59 recorded interaction occurrences / 56 distinct historical IDs**.
- Chronological ID range: [AI-001](#) through [AI-056](#); no numeric ID in that range is missing.
- P14 itself is appended as [AI-057](#) after the historical corpus is compiled.
- Full historical entries are preserved below; P14 does **not** recreate prompts, outputs, human decisions, execution results, screenshots, or defects that were not already recorded.
- PDF export is intentionally deferred because [plan.md](#) states [Export PDF later](#).

Required-field normalization

The historical entries were written incrementally while the assignment was being performed, so some labels differ while their semantics are equivalent. P14 uses the following mapping without rewriting the original interaction text:

Phase-8 required field	Historical audit representation
Tool	AI Tool
Date	Date portion of Date and Time
Time	Time/offset portion of Date and Time
Prompt	User Prompt, User Decision / Prompt, User Decision, or User Input
Output	AI Output
Human Review	Human Review, Human Review / Correction, Human Review / Finalization, and/or a Human Review Addendum
Correction	Recorded only where a human modification, testcase correction, re-review, parser fix, or other correction actually occurred; absence does not imply a fabricated correction
Artifact	Generated/Modified Files plus any explicitly cited external record/evidence

Validation of the 59 pre-P14 historical occurrences found:

- Tool present: **59/59**
- Date and Time present: **59/59**
- Prompt/decision/input present through the mapping above: **59/59**
- AI Output present: **59/59**
- Artifact field present: **59/59**
- Human Review section/addendum present: **59/59**
- Explicit correction signal recorded where applicable: **41/59**; the remaining entries are preserved without inventing a correction event.

Historical numbering anomalies preserved for traceability

Two numbering mistakes occurred while the audit was being appended incrementally:

- **AI-010** appears **three times**, all for successive FR-05 Step-F generation interactions at **14:40**, **15:00**, and **15:06** on 2026-08-20.
- **AI-047** appears **twice**, for FR-16 Step I at **11:09:14** and Step J at **11:17** on 2026-08-21.

These headings are intentionally **not renumbered retroactively** because other artifacts or human review notes may already refer to those IDs. Date/time + stage provide disambiguation. The next new entry uses the next unused ID, **AI-057**.

Coverage of the historical interaction corpus

Area	Recorded interaction occurrences
Setup / shared workflow	6
FR-05 pipeline	20
FR-10 pipeline	12
FR-16 pipeline	16
P9–P13 consolidation / CI / generator / diagram	5

Area	Recorded interaction occurrences			
Total before P14			59	
Human-AI collaboration evidence				
The report demonstrates the required pattern AI Output → Human Evaluation → Correction → Final Artifact using actual recorded interactions rather than reconstructed examples:				
Evidence chain	AI Output	Human Evaluation	Correction / decision	Final artifact/result
AI-002	Initial API-testing skill contained HW06-specific constraints	Human requested a reusable, project-neutral skill	Removed fixed FR/count/header/Postman assumptions and made project constraints runtime-configurable	.agents/skills/api-testing-human-loop/reusable skill
AI-005 → AI-006	FR-05 requirements/domain options preserved unsupported behavior as unresolved	Human explicitly chose Gate-A/Gate-B options	Search semantics, length, whitespace, duplicate-query behavior remained characterization/unresolved where required	Approved FR-05 requirement + domain models
AI-029 → AI-030	Six FR-10 AI cases remained INCOMPLETE after human audit	Human reviewed corrected versions	Added deterministic corrected tests, then human re-approved all six while preserving original audit provenance	42/42 FR-10 AI cases implementation-ready
AI-043 → AI-045	FR-16 human audit recorded 45 VALID / 7 INCOMPLETE	Human required resolution of the seven incomplete cases	Corrections were proposed, reviewed, then explicitly human-approved	Corrected FR-16 generated suite ready for implementation
AI-054	CI design needed both green and exactly-one-fail evidence despite real Step-K defects	Human assignment requirements required truthful separation	Built a dedicated CI-demo suite; intentional fail labelled CI-DEMO-ONLY and explicitly not a SUT defect	Real GitHub Actions pass/fail/restore evidence
AI-055	Generator initially parsed the 100 prefix of price: 10000 as HTTP 100	Validation exposed the parser error	Status regex changed to require a recognized HTTP reason phrase; duplicate body-field model fixed	Regression-passing API test generator
AI-056	AI supplied the earlier diagram blueprint/draft	Student manually	Student-reviewed Mermaid was promoted and re-	Final P13 .mmd + .png diagram artifacts

Evidence chain	AI Output	Human Evaluation	Correction / decision	Final artifact/result
		reviewed/edited Mermaid	rendered; SHA-256 matched the student's PNG	

Evidence-integrity rules applied during compilation

- No runtime PASS/FAIL was created by P14.
- No bug was newly claimed by P14.
- No screenshot, Newman report, GitHub Actions result, or user decision was fabricated.
- Official Step-K evidence remains the source of truth for API execution and confirmed defects.
- The CI intentional-failure sample remains explicitly separate from SUT defect evidence.
- Existing pending-review wording is retained when there is no later explicit human approval recorded in the audit itself.
- Only the two session-log files that actually exist are indexed in `reports/ai-session-logs/README.md`; missing historical transcripts are not reconstructed.

AI Tools Used

AI Tool	Tasks
ChatGPT	HW06 planning, skill design, requirement/domain/state/security/schema analysis support, testcase generation/correction support, Postman/Newman finalization, CI/CD integration, API-test-generator implementation, diagram review support, and report compilation.
Liebe local coding connector	Local repository inspection/editing, command execution, validation, evidence collection, Git/GitHub operations, and artifact finalization under ChatGPT direction.

New material AI interactions are appended chronologically using `$ai-audit-report`. Human review remains pending until explicitly completed by the tester.

AI Audit Entry - AI-001

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T10:23:00+07:00
Stage	P1 — AI Audit Skill refinement
API	N/A
User Prompt	Skill AI Audit của HW06 bạn có thể tham khảo các HW trước đó như <code>/ky_3/SoftwareTesting-Homework/HW05/.agents/skills/ai-audit-report</code> và chỉnh lại cho phù hợp. Bạn không dùng lệnh <code>\$skill-creator</code> để tạo skill được à?
Evidence/Input Basis	<code>HW05/.agents/skills/ai-audit-report/SKILL.md</code> ; <code>HW05/.agents/skills/ai-audit-report/references/audit-entry-template.md</code> ; <code>HW05/.agents/skills/ai-audit-report/agents/openai.yaml</code> ; <code>HW06/plan.md</code> ; local system skill <code>~/.codex/skills/.system/skill-creator/SKILL.md</code>

Field	Content
Generated/Modified Files	HW06/.agents/skills/ai-audit-report/SKILL.md; HW06/.agents/skills/ai-audit-report/agents/openai.yaml; HW06/.agents/skills/ai-audit-report/references/audit-entry-template.md; HW06/reports/ai-audit-report.md; HW06/reports/ai-session-logs/.gitkeep; HW06/plan.md; removed the earlier custom HW06/skills/ai-audit/ and HW06/ai-audit/ implementation
AI Output	Refactored P1 to follow the established HW05 audit-skill pattern and the system <code>\$skill-creator</code> guidance. The skill now uses a concise <code>SKILL.md</code> , OpenAI interface metadata, a reusable audit-entry reference template, chronological report appending, explicit pending human review, and HW06-specific evidence-integrity rules. The system validator was invoked but could not complete because local Python lacks the <code>yaml</code> module; no dependency was installed automatically.

Human Review:

- Status: Pending human review
- Accepted:
- Modified:
- Removed:
- Added:
- Notes:

AI Audit Entry - AI-002

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T10:30:00+07:00
Stage	P2 — API Testing Human-in-the-Loop Skill creation
API	N/A — shared workflow for FR-05, FR-10, FR-16
User Prompt	Bước tiếp theo là P2 — tạo <code>API Testing Human-in-the-Loop Skill</code>, và lần này tôi sẽ dùng ngay <code>\$skill-creator</code> pattern. Lưu ý: Không commit, tôi sẽ kiểm tra sau đó commit sau
Evidence/Input Basis	HW06/plan.md; HW05/.agents/skills/design-jmeter-load-test/SKILL.md; HW05/.agents/skills/design-jmeter-load-test/agents/openai.yaml; ~/.codex/skills/.system/skill-creator/SKILL.md; ~/.codex/skills/.system/skill-creator/references/openai_yaml.md
Generated/Modified Files	HW06/.agents/skills/api-testing-human-loop/SKILL.md; HW06/.agents/skills/api-testing-human-loop/agents/openai.yaml; HW06/.agents/skills/api-testing-human-loop/references/workflow-gates.md; HW06/.agents/skills/api-testing-human-loop/references/testcase-evidence-contract.md; HW06/plan.md; HW06/reports/ai-audit-report.md

Field	Content
AI Output	Created the P2 skill using the system <code>\$skill-creator</code> initializer with a references resource directory, then replaced the scaffold with HW06-specific Human-in-the-Loop instructions. Added explicit Gate A–H semantics, testcase/evidence provenance rules, protection for the required human-added testcases, real-execution evidence constraints, and integration with <code>\$ai-audit-report</code> . Fixed an initializer command shell-expansion issue in <code>agents/openai.yaml</code> . Manual structure validation passed. The official <code>quick_validate.py</code> was attempted but could not run because the local Python environment lacks the <code>yaml</code> module; no package was installed. No files were staged and no commit was created.

Human Review:

- Status: Pending human review
- Accepted:
- Modified:
- Removed:
- Added:
- Notes:

Human Review Addendum for AI-002 — 2026-08-20T10:48:00+07:00

- Status: Modified after human review
- Accepted: Use of `$skill-creator`, Human-in-the-Loop gates, contract fidelity, evidence integrity, and reusable references.
- Modified: Removed HW06-specific API assignments, testcase-count requirements, SEC-01–SEC-07 assumptions, `X-Student-Id`, fixed Postman/Newman requirements, and HW06-specific folder behavior from the skill. Project-specific constraints are now supplied as runtime/task context.
- Removed: Fixed references to FR-05, FR-10, FR-16 and assignment-specific provenance/count rules as universal skill behavior.
- Added: Tool/framework neutrality, generic contract extraction, reusable state/security/schema coverage guidance, and configurable project-specific constraints.
- Notes: The skill remains stored inside HW06 for this project, but its instructions are intentionally reusable for future API-testing work.

AI Audit Entry - AI-003

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T11:24:00+07:00
Stage	Phase 2 — Obtain and Inspect SUT
API	Full SUT inventory; focus later on FR-05, FR-10, FR-16
User Prompt	Tôi đã thực hiện clone document/eshop-sut dựa vào đó thực hiện phase 2 Obtain and Inspect SUT
Evidence/Input Basis	<code>/Users/nguyenhoangphihung/Document/eshop-sut/README.md</code> ; <code>/Users/nguyenhoangphihung/Document/eshop-</code>

Field	Content
	sut/api_specification.md; /Users/nguyenhoangphihung/Document/eshop-sut/backend/server.js; repository metadata at commit 85af3ba875c88283615e22cb108f13e2fccaf0e9; HW06/plan.md
Generated/Modified Files	HW06/source/api_specification.md; HW06/reports/api-inventory.md; HW06/reports/ai-audit-report.md
AI Output	Verified the local SUT clone and repository snapshot, copied the API specification into the HW06 source folder, inventoried 30 documented routes and 31 implemented routes, mapped FR/security/state/validation concerns, separated business requirements from API contract and implementation observations, and recorded unresolved/conflicting contract points. Key conflict: FR-16 business requirements describe CSV upload/RFC4180/atomic rollback while the API specification and implementation use a JSON products array. Static mismatch candidates were recorded but explicitly not classified as confirmed bugs without real execution. No runtime API tests were executed and no commit was created.

Human Review:

- Status: Pending human review
- Accepted:
- Modified:
- Removed:
- Added:
- Notes:

Human Review Addendum for AI-003 — 2026-08-20T13:32:00+07:00

- Status: Accepted with source-handling decision
- Accepted: Phase 2 inventory, conflict preservation, FR/SEC/state mapping, and separation of static observations from confirmed defects.
- Modified: Adopt Option B as the authoritative source-handling rule for later phases.
- Removed: None.
- Added: README.md = business truth; api_specification.md = API interface contract; implementation/runtime = actual behavior. Later tests must distinguish Business Compliance Tests from API Contract Tests and use **CONFIRMED DEFECT** only after real execution plus human confirmation.
- Notes: Gate A source-precedence decision is approved. Existing unresolved response/status/schema items remain unresolved unless supported by requirement/contract evidence.

AI Audit Entry - AI-004

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T13:35:00+07:00
Stage	Phase 3 — Select Three APIs
API	FR-05 Product Search; FR-10 Cancel Order; FR-16 Import Products

Field	Content
User Prompt	Tiếp tục với phase 3
Evidence/Input Basis	HW06/plan.md; HW06/reports/api-inventory.md; HW06/source/api_specification.md; /Users/nguyenhoangphihung/Document/eshop-sut/README.md; /Users/nguyenhoangphihung/Document/eshop-sut/backend/server.js; Human Gate A Option B decision
Generated/Modified Files	HW06/reports/api-selection.md; HW06/reports/ai-audit-report.md
AI Output	Verified the already-assigned API combination instead of performing a new selection. Confirmed one API from each required pool, verified each endpoint exists in both API specification and implementation, documented Requirement/API Contract/Conflict/Implementation Observation layers under Option B, and recorded why FR-05, FR-10, and FR-16 provide complementary testing dimensions. Did not fabricate proof that no group member has the same combination; that remains a human/group coordination assertion. No runtime execution and no commit were performed.

Human Review:

- Status: Pending human review
- Accepted:
- Modified:
- Removed:
- Added:
- Notes:

AI Audit Entry - AI-005

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T13:37:00+07:00
Stage	Phase 4 / API 1 / Step A — Requirement Extraction
API	FR-05 Product Search — GET /api/products?search=keyword
User Prompt	bắt đầu Phase 4 với API 1 — FR-05 Product Search , trước tiên là Step A — Requirement Extraction .
Evidence/Input Basis	/Users/nguyenhoangphihung/Document/eshop-sut/README.md FR-05 and SEC-04/SEC-05; /Users/nguyenhoangphihung/Document/eshop-sut/api_specification.md section 3.1; /Users/nguyenhoangphihung/Document/eshop-sut/backend/server.js product-search route; HW06/reports/api-inventory.md; approved Option B source model
Generated/Modified Files	HW06/PoolA-FR-05-ProductSearch/analysis/requirements.md; HW06/reports/ai-audit-report.md

Field	Content
AI Output	Extracted source-supported FR-05 business requirements, API contract, applicable security requirements, request inputs, response expectations, state applicability, and unresolved items. Explicitly separated UI-only requirements (loading, h1, presentation/empty-state rendering) from API-testable behavior. Preserved unspecified status codes, response schemas, case sensitivity, trimming, length limits, pagination and duplicate-query semantics as UNRESOLVED rather than inventing expectations. Recorded the SQL interpolation in backend/server.js as an implementation observation and potential SEC-05 mismatch, not a confirmed defect. No runtime API execution and no commit were performed.

Human Review:

- Status: Pending human review
- Accepted:
- Modified:
- Removed:
- Added:
- Notes:

Human Review Addendum for AI-005 — 2026-08-20T14:12:00+07:00

- Status: Accepted with explicit Step A decisions
- Accepted: FR-05 requirement extraction, Option B source separation, API-vs-UI scoping, and preservation of unsupported expectations as unresolved.
- Modified: Human selected 1C, 2C, 3C, 4C, 5C, 6C, 7C, 8B for the eight unresolved review items.
- Removed: None.
- Added: Case sensitivity, whitespace, search length, no-result payload/status, success status, and full schema remain **UNRESOLVED**; pagination/sort/filter is **NOT SPECIFIED / NOT IN SCOPE**; duplicate **search** parameters may be tested only as robustness/characterization behavior and cannot alone establish FR-05 non-compliance.
- Notes: Human Gate A for API 1 Step A is complete. Proceed to Step B — Domain Partition Design using these constraints.

AI Audit Entry - AI-006

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T14:20:00+07:00
Stage	Phase 4 / API 1 / Step B — Domain Partition Design
API	FR-05 Product Search — GET /api/products?search=keyword
User Prompt	Using \$api-testing-human-loop để thực hiện Step B
Evidence/Input Basis	HW06/.agents/skills/api-testing-human-loop/SKILL.md ; HW06/.agents/skills/api-testing-human-loop/references/workflow-gates.md ; approved Step A decisions in PoolA-FR-05-

Field	Content
	ProductSearch/analysis/requirements.md ; HW06/plan.md ; existing PoolA-FR-05-ProductSearch/analysis/domain-partitions.md
Generated/Modified Files	HW06/PoolA-FR-05-ProductSearch/analysis/domain-partitions.md ; HW06/reports/ai-audit-report.md
AI Output	Applied \$api-testing-human-loop Gate B rules to review the existing FR-05 domain model. Confirmed the only documented parameter search is covered without inventing typed null, numeric boundary, or length constraints. Preserved controlled-data requirements, SEC-05 input-domain coverage, and Step A unresolved decisions. Added explicit Human Gate B review options for exact-vs-partial matching, empty/whitespace partition granularity, security-oriented input placement, long-input representatives, and duplicate-query handling. No human option was selected by the AI. No runtime API execution, staging, or commit was performed.

Human Review:

- Status: Pending human review
- Accepted:
- Modified:
- Removed:
- Added:
- Notes:

Human Review Addendum for AI-006 — 2026-08-20T14:23:00+07:00

- Status: Accepted with explicit Gate B decisions
- Accepted: Domain partition coverage, non-invention of unsupported boundaries/types, controlled-data requirements, and separation of characterization from contractual expectations.
- Modified: Human selected 1B, 2B, 3B, 4B, 5B.
- Removed: None.
- Added: Partial-name search remains characterization; empty/whitespace/padded-whitespace stay separate; security-oriented values remain input-domain classes with detailed security analysis deferred to Step D; long keywords remain robustness representatives without contractual max-length assertions; duplicate search parameters remain robustness/characterization only.
- Notes: Human Gate B is complete. Step B partition model is approved and ready for Step C — State/Sequence Analysis.

AI Audit Entry - AI-007

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T14:26:00+07:00
Stage	Phase 4 / API 1 / Step C — State/Sequence Analysis
API	FR-05 Product Search — GET /api/products?search=keyword
User Prompt	Tiếp tục sử dụng \$api-testing-human-loop cho Step C — State/Sequence Analysis

Field	Content
Evidence/Input Basis	HW06/.agents/skills/api-testing-human-loop/SKILL.md; HW06/.agents/skills/api-testing-human-loop/references/workflow-gates.md; approved Step A requirements; approved Step B domain partitions; HW06/plan.md Step C
Generated/Modified Files	HW06/PoolA-FR-05-ProductSearch/analysis/state-transitions.md; HW06/reports/ai-audit-report.md
AI Output	Applied \$api-testing-human-loop Gate C. Determined that FR-05 has no explicit business state machine and did not fabricate lifecycle states. Modeled dataset conditions as preconditions, request independence/repeatability sequences, public-auth context, and concurrency only as a data-stability concern. Added explicit Human Gate C options for whether to model a formal state machine, repeatability expectations, cross-query sequences, authentication context, and concurrency treatment. No human option was selected by the AI. No runtime execution, staging, or commit was performed.

Human Review:

- Status: Pending human review
- Accepted:
- Modified:
- Removed:
- Added:
- Notes:

Human Review Addendum for AI-007 — 2026-08-20T14:27:00+07:00

- Status: Accepted with explicit Gate C decisions
- Accepted: No explicit FR-05 business state machine, contextual dataset preconditions, read-only request sequences, and separation of sequence analysis from security/concurrency requirements.
- Modified: Human selected 1B, 2B, 3A, 4B, 5B.
- Removed: None.
- Added: Keep contextual states/sequences instead of artificial lifecycle states; repeatability is non-mutation plus equivalent semantics under stable data; retain Search A → Search B and Search → listing independence checks; authentication remains contextual and moves to Step D for detailed security analysis; concurrent dataset changes remain an environment/data-stability concern only.
- Notes: Human Gate C is complete. Step C is approved and ready for Step D — Security Analysis.

AI Audit Entry - AI-008

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T14:30:00+07:00
Stage	Phase 4 / API 1 / Step D — Security Analysis
API	FR-05 Product Search — GET /api/products?search=keyword
User Prompt	Tiếp tục \$api-testing-human-loop thực hiện step D

Field	Content
Evidence/Input Basis	HW06/.agents/skills/api-testing-human-loop/references/workflow-gates.md Gate D; approved Step A/B/C artifacts; HW06/plan.md Step D; /Users/nguyenhoangphihung/Document/eshop-sut/README.md SEC-01→SEC-07; /Users/nguyenhoangphihung/Document/eshop-sut/backend/server.js search route
Generated/Modified Files	HW06/PoolA-FR-05-ProductSearch/analysis/security.md; HW06/reports/ai-audit-report.md
AI Output	Applied \$api-testing-human-loop Gate D. Mapped SEC-01 through SEC-07 by actual applicability, with SEC-05 directly applicable and SEC-04 partially applicable. Marked role escalation, IDOR/BOLA, mass assignment, replay, and unrelated auth requirements as not applicable rather than forcing generic security cases. Modeled SQL-injection payload families, public-access context, invalid-token characterization, information-disclosure risk, and rate/abuse as not specified. Recorded direct SQL interpolation and HTML DB-error output as implementation observations, not confirmed defects. Added six Human Gate D review options and did not select them on the user's behalf. No runtime execution, staging, or commit was performed.

Human Review:

- Status: Pending human review
- Accepted:
- Modified:
- Removed:
- Added:
- Notes:

Human Review Addendum for AI-008 — 2026-08-20T14:33:00+07:00

- Status: Accepted with explicit Gate D decisions
- Accepted: SEC-05 as the primary API security requirement, SEC-04 as partial API coverage, non-applicable security categories excluded, and implementation observations kept separate from confirmed defects.
- Modified: Human selected 1B, 2A, 3B, 4B, 5B, 6B.
- Removed: None.
- Added: SEC-05 defect criterion requires unsafe/non-parameterized query behavior or input altering/breaking SQL structure; information disclosure remains a separate risk dimension; SEC-04 requires UI/E2E evidence for full verification; invalid/expired JWT is optional characterization on this public endpoint; rate/abuse is not specified; testcase generation is limited to applicable/partial/risk-based security dimensions.
- Notes: Human Gate D is complete. Step D is approved and ready for the Schema/Response Analysis checkpoint.

AI Audit Entry - AI-009

Field	Content
AI Tool	ChatGPT with Liebe local coding connector

Field	Content
Date and Time	2026-08-20T14:35:00+07:00
Stage	Phase 4 / API 1 / Step E — Schema Validation Design
API	FR-05 Product Search — <code>GET /api/products?search=keyword</code>
User Prompt	Tiếp tục \$api-testing-human-loop thực hiện Step E
Evidence/Input Basis	HW06/.agents/skills/api-testing-human-loop/references/workflow-gates.md Schema/Response checkpoint; approved Step A/B/C/D artifacts; HW06/plan.md Step E; /Users/nguyenhoangphihung/Document/eshop-sut/api_specification.md product sections; /Users/nguyenhoangphihung/Document/eshop-sut/README.md FR-05; /Users/nguyenhoangphihung/Document/eshop-sut/backend/server.js GET product-search route
Generated/Modified Files	HW06/PoolA-FR-05-ProductSearch/analysis/schema.md; HW06/reports/ai-audit-report.md
AI Output	Applied \$api-testing-human-loop Schema/Response checkpoint. Determined that GET <code>/api/products</code> has no formally specified response schema/status/error schema. Separated source-backed search semantics from characterization-only observations, avoided promoting CRUD request fields or implementation DB-row output into a GET response contract, and preserved no-match/status/type/additional-property details as unresolved. Added seven Human Review options covering top-level shape, product fields, primitive types, no-match representation, extra properties, error responses, and characterization-test inclusion. No human option was selected by the AI. No runtime execution, staging, or commit was performed.

Human Review:

- Status: Pending human review
- Accepted:
- Modified:
- Removed:
- Added:
- Notes:

Human Review Addendum for AI-009 — 2026-08-20T14:39:00+07:00

- Status: Accepted with explicit Schema/Response checkpoint decisions
- Accepted: Missing formal GET schema/status/error schema remains unresolved, while source-backed semantics and characterization checks stay separate.
- Modified: Human selected 1B, 2C, 3B, 4B, 5B, 6B, 7B.
- Removed: None.
- Added: Top-level response shape is characterization only; exact GET product fields and primitive types remain unresolved; no-match payload/status remains unresolved; extra properties do not fail by default; error schema remains unresolved with actual response recorded and information disclosure analyzed separately; Step F may include a limited number of clearly labeled CHARACTERIZATION schema cases.
- Notes: Schema/Response checkpoint is complete. Step E is approved and ready for Step F — AI Testcase Generation.

AI Audit Entry - AI-010

Field	Content
AI Tool	ChatGPT with <code>\$api-testing-human-loop</code> , Liebe local coding connector, and <code>artifact_tool</code> spreadsheet generation
Date and Time	2026-08-20T14:40:00+07:00
Stage	Phase 4 / API 1 / Step F — AI Testcase Generation
API	FR-05 Product Search — <code>GET /api/products?search=keyword</code>
User Prompt	Thực hiện tiếp tục Step F
Evidence/Input Basis	Approved API 1 Step A requirements; Step B domain partitions and Human Gate B decisions; Step C state/sequence model and Gate C decisions; Step D security analysis and Gate D decisions; Step E schema/response checkpoint decisions; <code>HW06/plan.md</code> Step F; <code>\$api-testing-human-loop</code> testcase/evidence contract
Generated/Modified Files	<code>HW06/PoolA-FR-05-ProductSearch/generated/ai-generated-tests.xlsx</code> ; <code>HW06/reports/ai-audit-report.md</code>
AI Output	Generated 42 AI-origin FR-05 testcases in seven staged batches rather than one generic batch: 6 happy-path/basic, 12 domain, 6 boundary/robustness, 5 state/sequence, 7 security, 4 schema, and 2 cross-check cases. Every testcase has <code>Source = AI</code> and includes <code>X-Student-Id: {{studentId}}</code> in its request input. Unsupported status codes and formal GET schemas remain <code>UNRESOLVED</code> ; characterization cases are labeled explicitly; human-audit and real-execution fields are left blank. Workbook coverage formulas verify 42 total cases and the ≥ 35 threshold. The first spreadsheet-engine startup attempt timed out before workbook creation; a fresh retry succeeded, and the final workbook was verified with no formula errors. No API requests were executed, no human audit classification was fabricated, no files were staged, and no commit was created.

Human Review:

- Status: Pending Human Gate E / Step G audit
- Accepted:
- Modified:
- Removed:
- Added:
- Notes:

AI Audit Entry - AI-010

Field	Content
AI Tool	ChatGPT with Liebe local coding connector + <code>artifact_tool</code> spreadsheet generation
Date and Time	2026-08-20T15:00:00+07:00
Stage	Phase 4 / API 1 / Step F — AI Testcase Generation
API	FR-05 Product Search — <code>GET /api/products?search=keyword</code>

Field	Content
User Prompt	Thực hiện tiếp tục Step F
Evidence/Input Basis	<code>\$api-testing-human-loop</code> testcase/evidence contract; approved Step A requirements; approved Step B domain partitions; approved Step C state/sequence analysis; approved Step D security analysis; approved Step E schema design; <code>HW06/plan.md</code> Step F and project constraints
Generated/Modified Files	<code>HW06/PoolA-FR-05-ProductSearch/generated/ai-generated-tests.xlsx</code> ; <code>HW06/reports/ai-audit-report.md</code>
AI Output	Generated 42 AI-origin FR-05 testcases in seven batches rather than one generic batch: 6 happy-path/basic, 12 domain, 6 boundary/negative, 5 state/sequence, 7 security, 4 schema, and 2 cross-check cases. Every testcase has <code>Source = AI</code> , includes the required <code>X-Student-Id={{studentId}}</code> request header in its input definition, traces to approved requirements/partitions/state/security/schema bases, and preserves unsupported status/schema/normalization expectations as <code>UNRESOLVED</code> or <code>CHARACTERIZATION</code> . Human Audit Status, Human Audit Reason, Execution Status, Actual Result, Evidence, and Defect ID are intentionally blank. Workbook coverage formulas verified Total=42 and Threshold met=YES, with no spreadsheet formula errors detected. No API execution, Newman evidence, human review classification, staging, or commit was performed.

Human Review:

- Status: Pending Step G human audit
- Accepted:
- Modified:
- Removed:
- Added:
- Notes: Step F generation is complete. The generated cases must next be audited individually with project labels VALID / INVALID / INCOMPLETE; AI must not fill those labels as human decisions.

AI Audit Entry - AI-010

Field	Content
AI Tool	ChatGPT with Liebe local coding connector + artifact_tool spreadsheet generation
Date and Time	2026-08-20T15:06:00+07:00
Stage	Phase 4 / API 1 / Step F — AI Testcase Generation
API	FR-05 Product Search — <code>GET /api/products?search=keyword</code>
User Prompt	Thực hiện tiếp tục Step F
Evidence/Input Basis	<code>\$api-testing-human-loop</code> testcase/evidence contract; approved Step A requirements; approved Step B partitions; approved Step C state/sequence model; approved Step D security model; approved Step E schema/response checkpoint; <code>HW06/plan.md</code> Step F and project header/count requirements
Generated/Modified Files	<code>HW06/PoolA-FR-05-ProductSearch/generated/ai-generated-tests.xlsx</code> ; <code>HW06/reports/ai-audit-report.md</code>

Field	Content
AI Output	Generated 42 AI-origin FR-05 testcases in seven batches rather than one generic generation pass. Every testcase has Source = AI and includes X-Student-Id: {{studentId}} in its request input. Coverage workbook reports: Functional 8, Domain 12, Boundary 6, State/Sequence 5, Security 7, Schema 4, Total 42, threshold >=35 met. Expected HTTP status remains UNRESOLVED where the contract does not define it. Schema/normalization/duplicate-query cases approved as characterization remain explicitly labeled as such. Human audit, execution, actual-result, evidence, and defect fields are intentionally blank. No API request was executed and no evidence was fabricated. No staging or commit was performed.

Human Review:

- Status: Pending human review / Gate E
- Accepted:
- Modified:
- Removed:
- Added:
- Notes: Step G must audit every AI testcase using VALID / INVALID / INCOMPLETE. This entry does not claim that review is complete.

AI Audit Entry - AI-011

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T15:25:00+07:00
Stage	Phase 4 / API 1 / Step F artifact conversion before Step G
API	FR-05 Product Search — GET /api/products?search=keyword
User Prompt	chuyển bộ 42 testcase hiện tại từ XLSX sang Markdown và dùng Markdown làm source chính từ Step G trở đi
Evidence/Input Basis	Existing PoolA-FR-05-ProductSearch/generated/ai-generated-tests.xlsx containing the 42 Step F AI-generated testcases
Generated/Modified Files	HW06/PoolA-FR-05-ProductSearch/generated/ai-generated-tests.md ; HW06/reports/ai-audit-report.md
AI Output	Converted the existing 42 FR-05 AI-generated testcase records from XLSX into Markdown while preserving IDs AI-FR05-001 through AI-FR05-042 , Source = AI , requirement traces, UNRESOLVED/CHARACTERIZATION labels, and blank human-audit/execution/evidence/defect fields. The Markdown file is now explicitly designated as the review source of truth from Step G onward; the XLSX file is retained as an export/summary artifact only. Verification counted exactly 42 testcase sections and zero pre-populated Human Audit Status values. No testcase semantics were reclassified, no API execution was performed, and no staging or commit was performed.

Human Review:

- Status: User explicitly requested this artifact-source change
- Accepted: Markdown becomes the primary editable/review source from Step G onward
- Modified: XLSX role changed from primary Step F artifact to export/summary artifact
- Removed: None
- Added: PoolA-FR-05-ProductSearch/generated/ai-generated-tests.md
- Notes: Step G should read and update the Markdown file, preserving AI provenance. XLSX may be regenerated/synchronized later if needed for final submission.

Student ID Header Update — 2026-08-20T16:41:00+07:00

- User provided StudentID: 23127194.
- Updated the primary FR-05 Markdown testcase source so every testcase now carries X-Student-Id: 23127194 instead of the {{studentId}} placeholder.
- Normalized the testcase request notation to exact HTTP-header form: Header: X-Student-Id: 23127194.
- Newman/HTML report is not generated at this stage because Postman implementation and real execution have not occurred yet; no report or execution evidence was fabricated.

AI Audit Entry - AI-012

Field	Content
AI Tool	ChatGPT
Date and Time	2026-08-20T16:57:00+07:00
Stage	Phase 4 / API 1 / Step G — Human Audit revision review
API	FR-05 Product Search — GET /api/products?search=keyword
User Prompt	hãy thực hiện cập nhật phần audit tôi sẽ xem lại và đề xuất 5 test case khác
Evidence/Input Basis	User-uploaded PoolA-FR05-ProductSearch-test-AUDITED.md; previous Step A-E approvals; prior review feedback on AI-FR05-013, AI-FR05-035, AI-FR05-040 and the five human-added cases
Generated/Modified Files	Updated conversation artifact PoolA-FR05-ProductSearch-test-AUDIT-UPDATED.md; HW06/reports/ai-audit-report.md
AI Output	Revised the audit summary from 34 VALID / 8 INCOMPLETE to 32 VALID / 10 INCOMPLETE. AI-FR05-035 and AI-FR05-040 were changed to INCOMPLETE because their error-trigger inputs were not concrete/reproducible. AI-FR05-013 remains INCOMPLETE, but its reason now explicitly avoids inventing a mandatory Content-Type requirement and preserves the SEC-04 API-vs-UI boundary. Removed the previous HUM-FR05-001..005 set from the retained suite so the reviewer can propose five new genuinely human-authored cases. Current retained suite: 42 AI cases, 0 human-added cases pending replacement. No execution/evidence/defect fields were populated.

Human Review:

- Status: Pending reviewer re-check of the revised audit and pending five new human-authored cases
- Accepted:
- Modified:

- Removed: Previous HUM-FR05-001..005 from the retained suite
- Added: None — AI intentionally did not generate replacement HUMAN cases
- Notes: Human provenance requirement remains unsatisfied until the reviewer supplies five new cases.

AI Audit Entry - AI-013

Field	Content
AI Tool	ChatGPT
Date and Time	2026-08-20T17:05:00+07:00
Stage	Phase 4 / API 1 / Step H — Human-added testcase translation and tightening
API	FR-05 Product Search — GET /api/products?search=keyword plus documented product mutation APIs where applicable
User Prompt	Vậy hãy chuyển sang tiếng anh và cập nhật giúp tôi
Evidence/Input Basis	Six reviewer-authored HUMAN-FR05-043..048 proposals supplied directly in chat; previously approved Gate A-E decisions; audit decision to exclude unsupported draft/published lifecycle assumptions
Generated/Modified Files	Conversation artifact PoolA-FR05-ProductSearch-test-AUDIT-FINAL.md ; HW06/reports/ai-audit-report.md
AI Output	Translated and tightened the reviewer-authored human cases while preserving HUMAN provenance. Retained five cases: 043 cross-user header-context characterization, 044 concurrent search isolation, 046 rename→search state-history, 047 documented delete→search state-history, and 048 repeated expensive-search robustness. Removed 045 because draft/unpublished/published product lifecycle states are unsupported by the current source. Case 043 was constrained to characterization until X-Student-Id semantics are confirmed; 044 now uses a fixed reproducible workload; 046 removes any assumed eventual-consistency window; 047 removes hidden/soft-delete assumptions; 048 remains robustness/characterization because rate/abuse policy is NOT SPECIFIED. No execution evidence or defects were created.

Human Review:

- Status: Pending final human confirmation of the five retained HUMAN cases
- Accepted: Reviewer authorship/provenance of HUMAN-FR05-043, 044, 046, 047, 048
- Modified: English wording and scope tightened to remove unsupported assumptions
- Removed: HUMAN-FR05-045 from the retained set
- Added: None — no new HUMAN testcase invented by AI
- Notes: The downloadable conversation artifact is the updated review copy. The existing local project testcase source is not overwritten automatically by this entry; synchronize only after the reviewer accepts the five retained cases.

AI Audit Entry - AI-014

Field	Content
AI Tool	ChatGPT

Field	Content
Date and Time	2026-08-20T17:15:00+07:00
Stage	Phase 4 / API 1 / Step G correction pass before Postman implementation
API	FR-05 Product Search — <code>GET /api/products?search=keyword</code>
User Prompt	sửa 10 AI cases INCOMPLETE thành executable/reproducibl
Evidence/Input Basis	Final audited FR-05 testcase artifact plus the previously approved Gate A-E constraints and Human Audit reasons for AI-FR05-012, 013, 020, 029, 031, 032, 033, 034, 035, 040
Generated/Modified Files	<code>HW06/PoolA-FR-05-ProductSearch/generated/FR05-incomplete-corrections.md</code> ; downloadable corrected audit artifact <code>PoolA-FR05-ProductSearch-test-AUDIT-CORRECTED.md</code> ; <code>HW06/reports/ai-audit-report.md</code>
AI Output	Added concrete corrected inputs/steps/expectations for all ten previously INCOMPLETE AI cases while preserving the original Human Audit Status = INCOMPLETE labels. Corrections include: a fixed punctuation literal (012); explicit API-observation target without inventing JSON Content-Type contract (013); exactly 2,000 ASCII A characters for long-input robustness (020); exact reuse of AI-FR05-030 then AI-FR05-002 for the sequence case (029); fixed SQL-oriented literals for boolean, comment, UNION, and encoded mixed payloads (031-034); and a fixed unmatched-quote probe reused from AI-FR05-030 for the disclosure/error-schema cases (035, 040). Each corrected case is marked PENDING HUMAN RE-REVIEW ; AI did not reclassify any Human Audit decision to VALID . No API execution, Newman evidence, or defects were fabricated.

Human Review:

- Status: Pending human re-review of the ten corrected AI cases
- Accepted:
- Modified:
- Removed: None
- Added: Corrected executable/reproducible versions only; original AI records and audit labels remain preserved
- Notes: After reviewer approval, these corrected versions can be used as the implementation basis for Postman/Newman.

AI Audit Entry - AI-015

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T17:22:00+07:00
Stage	Phase 4 / API 1 / Step G — Human re-review approval
API	FR-05 Product Search — <code>GET /api/products?search=keyword</code>
User Prompt	Tôi đã kiểm tra qua và 10 AI case đó đã valid bạn hãy cập nhật giúp tôi

Field	Content
Evidence/Input Basis	Reviewer explicitly approved the corrected versions of AI-FR05-012, 013, 020, 029, 031, 032, 033, 034, 035, 040 as VALID
Generated/Modified Files	HW06/PoolA-FR-05-ProductSearch/generated/FR05-incomplete-corrections.md; HW06/reports/ai-audit-report.md
AI Output	Updated all ten corrected AI cases from PENDING HUMAN RE-REVIEW to VALID – HUMAN RE-REVIEW APPROVED . The original audit history is preserved, but the corrected versions are now the approved implementation basis. As a result, the final human audit outcome for the 42 AI-generated FR-05 cases is 42 VALID / 0 INCOMPLETE / 0 INVALID. No testcase content, execution status, evidence, or defect data was fabricated or changed.

Human Review:

- Status: APPROVED
- Accepted: AI-FR05-012, 013, 020, 029, 031, 032, 033, 034, 035, 040 corrected versions
- Modified: Re-review status only
- Removed: None
- Added: None
- Notes: Step G human audit is now complete for all 42 AI-generated FR-05 testcases.

AI Audit Entry - AI-016

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T17:25:00+07:00
Stage	Phase 4 / API 1 / Step H — Human-added testcase approval
API	FR-05 Product Search — GET /api/products?search=keyword plus documented product mutation APIs where applicable
User Prompt	5 Human added case đã được tôi xem qua và approve
Evidence/Input Basis	Reviewer explicitly approved the retained human-authored cases HUMAN-FR05-043 , HUMAN-FR05-044 , HUMAN-FR05-046 , HUMAN-FR05-047 , HUMAN-FR05-048
Generated/Modified Files	HW06/reports/ai-audit-report.md
AI Output	Recorded final human approval of the five retained reviewer-authored FR-05 testcases. Their provenance remains Source = HUMAN ; no testcase content was generated or relabeled by AI. Together with the completed audit of 42 AI-generated cases, the FR-05 suite is now finalized at 47 retained testcases: 42 AI cases approved VALID and 5 genuinely human-authored cases approved by the reviewer. No execution status, evidence, Newman result, or defect data was created.

Human Review:

- Status: APPROVED

- Accepted: HUMAN-FR05-043, HUMAN-FR05-044, HUMAN-FR05-046, HUMAN-FR05-047, HUMAN-FR05-048
- Modified: None
- Removed: HUMAN-FR05-045 remains excluded from the retained suite
- Added: None
- Notes: Step H / Human Gate F is complete for FR-05. The suite is ready for Postman implementation and subsequent real Newman execution.

AI Audit Entry - AI-017

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T18:01:00+07:00
Stage	Phase 4 / API 1 / Human-added testcase replacement before Postman execution
API	FR-05 Product Search plus one malformed POST sequence against the product collection endpoint
User Prompt	Replace HUMAN-FR05-043, HUMAN-FR05-044, and HUMAN-FR05-048 with three new reviewer-authored cases, translate them to English, and update PoolA-FR05-ProductSearch-test.md .
Evidence/Input Basis	Three replacement testcases authored directly by the reviewer in chat; previously approved HUMAN-FR05-046 and HUMAN-FR05-047 remain retained.
Generated/Modified Files	HW06/PoolA-FR-05-ProductSearch/generated/PoolA-FR05-ProductSearch-test.md ; HW06/reports/ai-audit-report.md
AI Output	Replaced HUMAN-FR05-043 with structured query parameter injection using search[\$ne] ; replaced HUMAN-FR05-044 with an encoded null-byte parser-differential sequence; replaced HUMAN-FR05-048 with a deterministic GET→malformed unauthenticated POST→GET mutation-safety sequence. All three were translated to English while preserving Source = HUMAN , the mandatory X-Student-Id: 23127194 , and empty execution/evidence fields. HUMAN-FR05-046 and 047 were left semantically unchanged and their already-approved Human Audit Status was synchronized to VALID. The retained human-added set remains five cases: 043, 044, 046, 047, 048.

Human Review:

- Status: APPROVED / reviewer-authored replacements received
- Accepted: HUMAN-FR05-043, HUMAN-FR05-044, HUMAN-FR05-048 replacement definitions; HUMAN-FR05-046 and HUMAN-FR05-047 remain approved
- Modified: English translation and formatting only for the three replacement definitions
- Removed: Previous versions of HUMAN-FR05-043, 044, 048
- Added: No AI-authored HUMAN cases
- Notes: Postman implementation must now be synchronized to these replacement definitions before the official Newman run.

AI Audit Entry - AI-018

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T18:08:00+07:00
Stage	HW06 workflow refinement — runtime-data preparation stage
API	Applies to FR-05, FR-10, and FR-16
User Prompt	Vậy hãy bổ sung step này vào plan và skill \$api-testing-human-loop
Evidence/Input Basis	Existing plan.md ; .agents/skills/api-testing-human-loop/SKILL.md ; workflow gate reference; prior decision that runtime test-data preparation is required for all three APIs but must be API-specific
Generated/Modified Files	HW06/plan.md ; HW06/.agents/skills/api-testing-human-loop/SKILL.md ; HW06/.agents/skills/api-testing-human-loop/references/workflow-gates.md ; HW06/reports/ai-audit-report.md
AI Output	Added a mandatory runtime-data/preconditions stage between test implementation and official execution. Plan Step J now prepares real runtime data, auth, IDs, states, fixtures, setup, cleanup, and readiness for all three APIs with API-specific examples for FR-05 search data, FR-10 controlled order states/actors, and FR-16 import fixtures/roles. Previous execution moved to Step K and bug reporting to Step L. The reusable skill and workflow-gates reference now prohibit starting official evidence execution until required preconditions are real and reproducible, and define READY/BLOCKED readiness states. No runtime data or execution evidence was fabricated.

Human Review:

- Status: REQUESTED / workflow update applied
- Accepted: Runtime test data and preconditions preparation as a reusable stage for all three APIs
- Modified: Plan step numbering and reusable skill workflow
- Removed: None
- Added: Runtime-data readiness checkpoint before official execution
- Notes: No files were staged or committed by this update.

AI Audit Entry - AI-019

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T18:24:00+07:00
Stage	API 1 / Step J — Prepare Runtime Test Data and Preconditions
API	FR-05 Product Search — GET /api/products?search=keyword
User Prompt	Bây giờ, bạn hãy thực hiện step J giúp tôi
Evidence/Input Basis	Live SUT on http://localhost:3000 ; current SQLite dataset; documented login/product endpoints; finalized FR-05 testcase source; current Postman environment/data/collection

Field	Content
Generated/Modified Files	postman/environment/HW06-FR05-Local.postman_environment.json; git-ignored postman/environment/HW06-FR05-Local.private.postman_environment.json; postman/data/FR05-runtime-data.json; postman/data/README.md; postman/data/FR05-runtime-readiness.md; .gitignore; reports/ai-audit-report.md
AI Output	Verified the live FR-05 SUT at http://localhost:3000 and discovered that 127.0.0.1:3000 currently maps to another local Node process returning 404 for /api/products . Verified the 5 seeded products and selected real controlled search values for matching, no-match, exact, partial, Unicode, case, one-character, A/B, and non-name-only scenarios. Performed a real admin login with the mandatory student header and verified the token maps to role admin ; stored the JWT only in a git-ignored private Postman environment. Created two disposable products through the product API for HUMAN-FR05-046 and HUMAN-FR05-047, IDs 6 and 7, and populated all non-secret mutation variables. Marked the 42 AI cases plus HUMAN-FR05-046/047 runtime-ready. Marked HUMAN-FR05-043/044/048 blocked because the current Postman collection still contains their superseded implementations. No official Newman execution, PASS/FAIL evidence, screenshot, or defect was produced.

Human Review:

- Status: Step J runtime-data preparation completed; collection synchronization remains required before Step K
- Accepted: Real runtime values, admin actor/token handling, disposable fixtures, readiness classification
- Modified: Runtime/environment data and documentation
- Removed: Old Postman limitation wording for superseded HUMAN cases
- Added: FR-05 runtime-readiness report and private token environment policy
- Notes: Official execution remains blocked until HUMAN-FR05-043/044/048 are reimplemented in the Postman collection.

AI Audit Entry - AI-020

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T21:42:00+07:00
Stage	API 1 / Human Gate G — Review real execution failures
API	FR-05 Product Search plus documented product-create mutation used by HUMAN-FR05-048
User Prompt	Review each failure against testcase expectation, README/business requirements, api_specification.md, and actual request/response evidence; determine the source basis for HUMAN-FR05-044 and HUMAN-FR05-048; only then approve confirmed defects and keep ID 8 until review finishes.
Evidence/Input Basis	Final HUMAN testcase source; SUT README from repository HEAD; source/api_specification.md ; official Newman execution summary; FR05-

Field	Content
	<code>official-failure-evidence.json</code> ; backend source inspected only for root-cause confirmation after execution evidence was established
Generated/Modified Files	<code>PoolA-FR-05-ProductSearch/evidence/FR05-human-gate-g-review.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Gate G review approved three contract-backed defect bases. HUMAN-FR05-044: the standalone expectations "no 5xx" and "no DB disclosure" are not directly specified for FR-05, but the observed SQLite failure plus backend inspection confirms direct search-input SQL interpolation, violating README SEC-05 parameterized-query requirement. HUMAN-FR05-048: <code>POST /api/products</code> is documented and therefore not an unsupported method; however README FR-12 requires valid JWT + admin role for all data-changing product APIs, and README FR-15 requires name, positive price, and valid category. The no-JWT malformed POST returned 200, created ID 8, and persisted null required fields, supporting two separate confirmed contract violations. ID 8 remains preserved for Step L evidence. No GitHub Issue or cleanup was performed.

Human Review:

- Status: APPROVED — Human Gate G complete
- Approved confirmed defect bases:
 1. FR05-BUG-01 — SEC-05 violation: product search concatenates user input into SQL instead of using a parameterized query.
 2. FR05-BUG-02 — FR-12 violation: unauthenticated client can create products through an admin-only data-changing product API.
 3. FR05-BUG-03 — FR-15 violation: product creation accepts missing required fields and persists an all-null product.
- Not promoted as standalone confirmed defects: HTTP 500 itself for unusual search input; SQLite-error disclosure itself, because no exact generic error-status/disclosure contract was found in the supplied requirements.
- Evidence state: product ID 8 intentionally retained until Step L documentation is complete.
- Notes: Step L may now create formal bug records; no staging or commit was performed here.

AI Audit Entry - AI-021

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T21:46:00+07:00
Stage	API 1 / Step L — Genuine Bug Reporting
API	FR-05 Product Search plus documented product mutation endpoint used by HUMAN-FR05-048
User Prompt	Chuyển sang Step L — Genuine Bug Reporting với 3 bug trên
Evidence/Input Basis	Human Gate G approved defect bases; official Newman evidence; testcase source; SUT README and api specification

Field	Content
Generated/Modified Files	<code>PoolA-FR-05-ProductSearch/bugs/FR05-BUG-01.md</code> ; <code>FR05-BUG-02.md</code> ; <code>FR05-BUG-03.md</code> ; <code>FR05-bug-summary.md</code> ; finalized testcase traceability in <code>generated/PoolA-FR05-ProductSearch-test.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Created three confirmed bug reports tied to SEC-05, FR-12, and FR-15 respectively. Severity assigned as High, Critical, High with rationale. Updated HUMAN-FR05-044 and HUMAN-FR05-048 execution/evidence/defect fields to point to the confirmed reports. Created a consolidated bug summary. Product ID 8 remains preserved for external screenshot/issue evidence. No GitHub Issue was fabricated or created because no authorized GitHub action was available/requested in this step. No cleanup, staging, or commit was performed.

Human Review:

- Status: Step L local bug documentation complete
- Confirmed bug records: FR05-BUG-01, FR05-BUG-02, FR05-BUG-03
- External issue/screenshot evidence: PENDING
- Cleanup: PENDING until external evidence is captured

AI Audit Entry - AI-022

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T22:10:00+07:00
Stage	API 1 / FR-05 — Post-evidence cleanup and final traceability
User Prompt	thực hiện luôn phần cleanup + update traceability cuối cho FR-05.
Evidence/Input Basis	User confirmed PNG evidence was attached to GitHub Issues #24/#25/#26; Step K official Newman evidence; Human Gate G; Step L bug reports and summary
Generated/Modified Files	<code>postman/data/FR05-runtime-data.json</code> ; <code>postman/environment/HW06-FR05-Local.postman_environment.json</code> ; git-ignored private environment; <code>postman/data/FR05-runtime-readiness.md</code> ; <code>postman/data/README.md</code> ; <code>PoolA-FR-05-ProductSearch/bugs/FR05-bug-summary.md</code> ; <code>PoolA-FR-05-ProductSearch/bugs/FR05-BUG-03.md</code> ; <code>PoolA-FR-05-ProductSearch/evidence/FR05-human-gate-g-review.md</code> ; <code>PoolA-FR-05-ProductSearch/generated/PoolA-FR05-ProductSearch-test.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Performed cleanup only after the user confirmed issue PNG attachments. Deleted unintended product ID 8, reset rename fixture ID 6 to <code>HW06_FR05_RENAME_23127194_OLD</code> , recreated delete fixture as ID 9, and synchronized runtime data/environments. Updated final traceability: FR05-BUG-01 → GitHub #24, FR05-BUG-02 → #25, FR05-BUG-03 → #26. Updated Step J/runtime notes, Gate G evidence-state note, bug summary, and BUG-03 evidence-state note to reflect completed evidence capture and cleanup. No Newman rerun, staging, or commit was performed.

Human Review:

- Status: COMPLETE — FR-05 cleanup and final traceability
- GitHub Issues: #24, #25, #26 with user-confirmed PNG attachments
- Cleanup: ID 8 deleted; fixture ID 6 reset; delete fixture recreated as ID 9
- Rerun readiness: restored
- Git staging/commit: not performed

AI Audit Entry - AI-023

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T22:25:00+07:00
Stage	API 1 / P1-P4 compliance cleanup
User Prompt	Thực hiện từ P1-P4 giúp tôi
Generated/Modified Files	PoolA-FR-05-ProductSearch/generated/PoolA-FR05-ProductSearch-test.md; PoolA-FR-05-ProductSearch/audit/audit-summary.md; PoolA-FR-05-ProductSearch/human-added/human-tests.md; PoolA-FR-05-ProductSearch/evidence/FR05-X-Student-Id-evidence.md; reports/ai-audit-report.md
AI Output	Synchronized stale master status with the completed official execution; preserved original 32 VALID / 10 INCOMPLETE audit history while marking the 10 corrected cases as explicitly human re-review approved; created a standalone audit summary; extracted the five HUMAN-authored cases into a dedicated Step H artifact and populated PASS evidence for the three non-defect HUMAN cases; verified from the official Newman JSON that all 62/62 executions carried X-Student-Id: 23127194. Attempted two real macOS Terminal screenshot capture paths; both were blocked by Screen Recording permissions, so no screenshot was fabricated.

Human Review:

- P1: COMPLETE
- P2: COMPLETE
- P3: COMPLETE
- P4 evidence verification: COMPLETE (62/62 official executions verified)
- P4 screenshot artifact: BLOCKED by macOS Screen Recording permission; manual capture still required unless permission is granted

AI Audit Entry - AI-024

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T22:42:00+07:00
Stage	Cross-API Step K optimization
User Prompt	Thực hiện khởi tạo và đưa nó vào workflow của step k

Field	Content
Generated/Modified Files	<code>scripts/run-step-k.sh</code> ; <code>scripts/extract-newman-summary.js</code> ; <code>scripts/README.md</code> ; <code>plan.md</code> ; <code>postman/newman/FR05-execution-summary.json</code> ; <code>postman/newman/FR05-execution-summary.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Added a reusable Step K runner supporting FR05/FR10/FR16 and a compact Newman summary extractor. Updated Step K so raw CLI/JSON/HTML/JUnit evidence is preserved on disk while AI reads the compact failure-oriented summary first. Validated the extractor against the existing FR05 official report: 47 testcase IDs, 62 requests, 142 assertions, 138 passed, 4 failed, failed cases HUMAN-FR05-044/HUMAN-FR05-048, and X-Student-Id 62/62. The raw FR05 JSON is about 1.09 MB while the generated Markdown summary is about 1.8 KB. FR05 runner dry-run resolves correctly; FR10 dry-run currently fails fast because its collection/environment/runtime-data artifacts do not yet exist. No live Newman rerun was performed, avoiding unnecessary mutation of the already-reviewed FR05 evidence state.

Human Review:

- Status: IMPLEMENTED AND VALIDATED
- Reuse scope: FR05, FR10, FR16
- Token-efficiency policy: compact summary first; raw report only for unresolved failed cases
- Evidence integrity: raw reports retained; Newman non-zero exit status preserved

AI Audit Entry - AI-025

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T22:55:00+07:00
Stage	API 2 / FR-10 — Human Gate A + Step B Domain Partition Design
API	<code>PUT /api/orders/:id/cancel</code>
User Decision	Accepted cross-user cancellation as a risk-based authorization expectation: User A must not cancel User B's order, without presenting it as direct FR-10 wording.
Generated/Modified Files	<code>PoolB-FR-10-CancelOrder/analysis/requirements.md</code> ; <code>PoolB-FR-10-CancelOrder/analysis/domain-partitions.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Marked Human Gate A approved and preserved the ownership/IDOR distinction as risk-based authorization. Designed Step B partitions for path <code>:id</code> , JWT/authentication, mandatory project header, undocumented request body/query shapes, content type, and state x ownership x auth combinations. No exact HTTP status codes or numeric ID boundaries were invented where the contract is silent. Semantic boundary <code>confirmed</code> →cancel allowed vs <code>shipping</code> →user cancel rejected is explicitly emphasized.

Human Review:

- Gate A: APPROVED
- Gate B: PENDING
- Ownership/IDOR provenance: RISK-BASED AUTHORIZATION EXPECTATION, not direct FR-10 wording

AI Audit Entry - AI-026

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T23:09:00+07:00
Stage	API 2 / FR-10 — Gate C approval + Step D Security Analysis
API	<code>PUT /api/orders/:id/cancel</code>
User Prompt	Thực hiện step D — Security Analysis.
Generated/Modified Files	<code>PoolB-FR-10-CancelOrder/analysis/state-transitions.md</code> ; <code>PoolB-FR-10-CancelOrder/analysis/security.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Marked Gate C approved and created FR-10 security analysis. Mapped SEC-01..SEC-07 with SEC-02 directly applicable and SEC-05 potentially applicable; preserved cross-user cancellation as mandatory risk-based BOLA/IDOR coverage rather than direct FR-10 wording. Added JWT negative cases, ownership isolation, state/status/user/id override attempts, selected injection/path probes, information-disclosure handling, fixture-isolation guidance, and optional concurrency/race characterization. No exact rejection status codes were invented where the contract is silent.

Human Review:

- Gate C: APPROVED
- Gate D: PENDING
- Concurrency: optional unless deterministic harness becomes available

AI Audit Entry - AI-027

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T23:13:00+07:00
Stage	API 2 / FR-10 — Gate D approval + Step E Schema/Response Analysis
API	<code>PUT /api/orders/:id/cancel</code>
User Prompt	bước tiếp theo đúng workflow là Step E — Schema/Response Analysis
Generated/Modified Files	<code>PoolB-FR-10-CancelOrder/analysis/security.md</code> ; <code>PoolB-FR-10-CancelOrder/analysis/schema.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Marked Gate D approved and created schema/response analysis for FR-10. The exact success/error HTTP codes and JSON schemas remain UNRESOLVED because the supplied contract does not define them. State transition/state preservation is used as the primary oracle, with follow-up <code>GET /api/orders/:id</code> recommended for

Field	Content
	before/after verification. Runtime response shape is treated as characterization unless later human review promotes it to a stable regression expectation.

Human Review:

- Gate D: APPROVED
- Schema/response checkpoint: PENDING
- Primary oracle: state transition/state preservation, not invented response schema

AI Audit Entry - AI-028

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T23:20:00+07:00
Stage	API 2 / FR-10 — Schema checkpoint approval + Step F AI Testcase Generation
API	PUT /api/orders/:id/cancel
User Prompt	Thực hiện Step F — AI Testcase Generation: tạo ≥35 AI-generated testcases cho FR10, dựa trên toàn bộ Requirement + Domain + State + Security + Schema vừa được approve.
Generated/Modified Files	PoolB-FR-10-CancelOrder/analysis/schema.md; PoolB-FR-10-CancelOrder/generated/PoolB-FR10-CancelOrder-test.md; reports/ai-audit-report.md
AI Output	Marked the schema/response checkpoint approved and generated 42 AI-provenance FR-10 testcases. Coverage: 4 Functional, 8 Domain, 6 Boundary, 9 State/Sequence, 11 Security, 4 Schema. All 42 preserve Source = AI, Human Audit Status = PENDING HUMAN REVIEW , blank execution evidence, and explicit UNRESOLVED/CHARACTERIZATION boundaries where the contract does not define exact HTTP status/body/schema. Cross-user cancellation remains labeled as Gate A-approved risk-based authorization coverage. Expired-token and concurrency cases were left outside the mandatory 42 because reproducible runtime setup/determinism is not yet established.

Human Review:

- Schema/response checkpoint: APPROVED
- Step F: COMPLETE — 42 AI-generated cases
- Step G human audit: NOT STARTED

AI Audit Entry - AI-029

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T23:43:00+07:00
Stage	API 2 / FR-10 — Step G human-audit localization + correction pass

Field	Content
API	<code>PUT /api/orders/:id/cancel</code>
User Input	Supplied completed human audit for all 42 AI-generated FR10 testcases: 36 VALID / 0 INVALID / 6 INCOMPLETE, with Human Audit Reason for every case.
Generated/Modified Files	<code>PoolB-FR-10-CancelOrder/generated/PoolB-FR10-CancelOrder-test.md; reports/ai-audit-report.md</code>
AI Output	Translated all Human Audit Reason entries into English while preserving the user's Human Audit Status decisions. Added a Step G audit summary and preserved the six original INCOMPLETE labels for provenance. Added deterministic Corrected Test content for AI-FR10-004, 011, 012, 017, 038, and 042: Newman JSON header evidence for 004; <code>curl --path-as-is</code> for 011; fixed <code>%27</code> path probe for 012; exact 4096-character safe payload for 017; fixed encoded SQL-style probe for 038; and suite-wide official Newman header verification bound to AI-FR10-001 for 042. No AI reclassification to VALID was performed; human re-review remains required for those six corrected cases.

Human Review:

- Original Step G audit: 36 VALID / 0 INVALID / 6 INCOMPLETE
- Human Audit Reason language: translated to English
- Corrections prepared: 6/6 INCOMPLETE cases
- Corrected-case re-review: PENDING HUMAN APPROVAL

AI Audit Entry - AI-030

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-20T23:58:00+07:00
Stage	API 2 / FR-10 — Human re-review approval for corrected AI cases
API	<code>PUT /api/orders/:id/cancel</code>
User Decision	Explicitly approved all six corrected AI testcases: AI-FR10-004, 011, 012, 017, 038, 042.
Generated/Modified Files	<code>PoolB-FR-10-CancelOrder/generated/PoolB-FR10-CancelOrder-test.md; reports/ai-audit-report.md</code>
AI Output	Preserved the six original Human Audit Status values as INCOMPLETE for provenance, marked each corrected version VALID — HUMAN RE-REVIEW APPROVED , and updated the Step G summary to state that all 42 AI-generated cases are implementation-ready after correction/re-review.

Human Review:

- Original audit: 36 VALID / 6 INCOMPLETE / 0 INVALID
- Corrected cases human re-reviewed: 6/6 APPROVED
- Final implementation-ready AI cases: 42/42

AI Audit Entry - AI-031

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T00:01:00+07:00
Stage	API 2 / FR-10 — Step H Human-authored Testcase Addition
API	PUT /api/orders/:id/cancel
User Input	Proposed six HUMAN-authored cases HUMAN-FR10-043..048 covering forged-body ownership bypass, foreign-vs-nonexistent enumeration, unauthorized-attempt history effects, JWT alg:none, duplicate JSON keys, and HTTP method override.
Generated/Modified Files	PoolB-FR-10-CancelOrder/human-added/human-tests.md; reports/ai-audit-report.md
AI Output	Reviewed all six cases and accepted them for Step H. Preserved Source=HUMAN provenance and refined wording for HUMAN-FR10-044 and HUMAN-FR10-048 so enumeration/method-override behavior is treated as risk-based security characterization rather than silently promoted to direct FR-10 contract requirements. Added execution-ready steps, expected state oracles, Why-AI-missed rationale, and Step H coverage mapping.

Human Review:

- HUMAN cases accepted: 6/6
- Required minimum: >=5
- Provenance: HUMAN
- Step H: COMPLETE

AI Audit Entry - AI-032

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T00:10:00+07:00
Stage	API 2 / FR-10 — Step I Postman Implementation
API	PUT /api/orders/:id/cancel
User Prompt	Thực hiện Step I
Generated/Modified Files	postman/collection/HW06-FR10-CancelOrder.postman_collection.json; postman/environment/HW06-FR10-Local.postman_environment.json; postman/environment/HW06-FR10-Local.private.postman_environment.json; PoolB-FR-10-CancelOrder/postman/implementation.md; reports/ai-audit-report.md
AI Output	Implemented 48 FR10 testcase IDs in Postman (42 AI + 6 HUMAN) using 53 executable requests because ordered-sequence cases require multiple requests. Added collection-level X-Student-Id: 23127194, semantic success/error-class assertions where contract-backed, documented GET /api/orders/:id follow-up

Field	Content
	state oracles, characterization logging for unresolved response schemas, corrected deterministic probes/payloads, and environment skeletons with runtime tokens/IDs intentionally blank until Step J. Preserved AI-FR10-011 raw-path correction as requiring <code>curl --path-as-is</code> evidence because Postman/Newman may normalize double slashes. Validation: 48/48 testcase IDs present, 57 scripts syntax-valid with 0 syntax errors, and Step K dry-run resolves collection/environment but correctly blocks only on missing <code>postman/data/FR10-runtime-data.json</code> .

Human Review:

- Step I: IMPLEMENTED
- Unique testcase IDs: 48
- Executable Postman requests: 53
- Runtime data readiness: NOT READY — Step J required
- Official Newman execution: NOT STARTED

AI Audit Entry - AI-033

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T00:12:00+07:00
Stage	API 2 / FR-10 — Step J Runtime Test Data and Preconditions
API	<code>PUT /api/orders/:id/cancel</code>
User Prompt	Thực step J — Prepare Runtime Test Data and Preconditions.
Generated/Modified Files	<code>scripts/prepare-fr10-runtime.py</code> ; <code>postman/data/FR10-runtime-data.json</code> ; git-ignored <code>postman/environment/HW06-FR10-Local.private.postman_environment.json</code> ; <code>PoolB-FR-10-CancelOrder/postman/runtime-fixture-manifest.json</code> ; <code>PoolB-FR-10-CancelOrder/postman/runtime-preconditions.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Verified the real EShop backend process on localhost:3000, created two dedicated timestamped regular users through register/login APIs, obtained the documented admin login, created 39 dedicated orders via <code>POST /api/checkout</code> , transitioned selected fixtures through the documented admin status endpoint, and verified the final ownership/state inventory. Runtime state counts: 27 pending, 7 confirmed, 3 shipping, 1 delivered, 1 canceled. User A owns 33 fixtures and User B owns 6. Updated the git-ignored private environment with real tokens/IDs, created non-secret FR10 runtime data and fixture manifest, added a reusable setup script using random fixture passwords for future runs, and documented reset/recreate strategy. Step K dry-run now resolves collection, private environment, runtime data, and Newman successfully. No official Newman run was performed.

Human Review:

- Step J runtime readiness: READY

- Dedicated orders: 39
- State/ownership verification mismatches: 0
- Private environment blank required variables: 0
- Step K dry-run: PASS
- Official execution: NOT STARTED

AI Audit Entry - AI-034

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T00:18:00+07:00
Stage	API 2 / FR-10 — Step I Postman Implementation
API	<code>PUT /api/orders/:id/cancel</code>
User Prompt	Thực hiện Step I.
Generated/Modified Files	<code>postman/collection/HW06-FR10-CancelOrder.postman_collection.json;</code> <code>postman/environment/HW06-FR10-Local.postman_environment.json;</code> <code>postman/environment/HW06-FR10-Local.private.postman_environment.json;</code> <code>PoolB-FR-10-CancelOrder/postman/implementation.md;</code> <code>reports/ai-audit-report.md</code>
AI Output	Implemented the 48 approved FR-10 testcase IDs in Postman: 42 AI + 6 HUMAN, represented by 53 executable requests because sequence cases require multiple requests. Added collection-level <code>X-Student-Id: 23127194</code> , environment variables, auth variants, state-oracle follow-up GET requests, corrected raw/body/security probes, response characterization logging, and HUMAN sequence/security coverage. JSON and 57 Postman scripts were syntax-validated with zero script syntax errors. <code>run-step-k.sh FR10 --dry-run</code> resolves successfully against the currently present runtime artifacts. No official Newman execution was performed as part of this Step I action.

Human Review:

- Step I implementation: COMPLETE
- Unique testcase IDs: 48 (42 AI + 6 HUMAN)
- Executable Postman requests: 53
- Postman script syntax errors: 0
- Official Newman execution in this action: NOT STARTED

AI Audit Entry - AI-035

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T08:44:00+07:00
Stage	API 2 / FR-10 — Step K Real Newman Execution + Gate G Classification
API	<code>PUT /api/orders/:id/cancel</code>

Field	Content
User Prompt	@Liebe Dùng \$api-testing-human-loop để thực hiện step K, nhớ sử dụng các script run-step-k.sh,... cho FR10
Generated/Modified Files	postman/newman/FR10-official-cli.txt; postman/newman/FR10-official-report.json; postman/newman/FR10-official-report.html; postman/newman/FR10-official-report.xml; postman/newman/FR10-execution-summary.json; postman/newman/FR10-execution-summary.md; PoolB-FR-10-CancelOrder/evidence/step-k-execution.md; scripts/prepare-fr10-runtime.py; postman/environment/HW06-FR10-Local.postman_environment.json; git-ignored private FR10 environment; postman/data/FR10-runtime-data.json; PoolB-FR-10-CancelOrder/postman/runtime-fixture-manifest.json; PoolB-FR-10-CancelOrder/postman/runtime-preconditions.md; FR10 Postman collection; reports/ai-audit-report.md
AI Output	Applied \$api-testing-human-loop Step K using the reusable runner. Dry-run resolved all inputs. The first real run produced 90 failed assertions because the existing order IDs were stale after an SUT reset; this was classified as runtime/setup failure, not a product bug. Recreated isolated runtime fixtures and reran, reducing failures to four testcase IDs. Gate G analysis identified AI-FR10-004 as a test implementation defect caused by sharing a mutation fixture with AI-FR10-001; added a dedicated canonical-request fixture, recreated the runtime dataset, and reran. Final current Newman evidence: 48 testcase IDs, 96 requests, 275 assertions, 269 passed, 6 failed, exact X-Student-Id coverage 96/96. Remaining failed IDs are AI-FR10-003, AI-FR10-021, and AI-FR10-040; all reproducibly show a User canceling a shipping order successfully and the order becoming canceled, contrary to FR-10. Static server inspection correlates with this behavior, but it remains a potential SUT defect until Human Gate H confirmation.

Human Review:

- Step K real execution: COMPLETE
- Gate G classification: PREPARED FOR HUMAN REVIEW
- Runtime/setup failure from stale fixtures: RESOLVED
- Postman implementation defect AI-FR10-004: RESOLVED AND RERUN PASS
- Remaining failed testcase IDs: 3
- Potential product defect: User can cancel shipping order
- Human Gate H: PENDING
- Step L bug report / GitHub Issue: NOT STARTED

AI Audit Entry - AI-036

Field	Content
AI Tool	ChatGPT with local coding connector
Date and Time	2026-08-21T08:59:55+07:00
Stage	API 2 / FR-10 — Step L confirmed-defect reporting after Human Gate H
API	PUT /api/orders/:id/cancel

Field	Content
User Decision	Explicitly approved Human Gate H: a User must not be able to cancel an order in shipping , while real Step K execution shows a successful cancellation and persisted canceled state.
Generated/Modified Files	PoolB-FR-10-CancelOrder/bugs/FR10-BUG-01.md ; reports/ai-audit-report.md
External Record	GitHub Issue #27: https://github.com/nhphunng/SoftwareTesting-Homework/issues/27
AI Output	Applied \$api-testing-human-loop at the confirmed-defect stage. Cross-checked FR-10 and the approved testcase definitions against the compact and raw real Newman evidence. Documented one High-severity defect traced to AI-FR10-003 , AI-FR10-021 , and AI-FR10-040 : the cancel request returned HTTP 200 and three isolated shipping orders were observed as canceled afterward. Recorded reproducibility as 3/3 independent testcase scenarios in the final controlled run, cited only existing Step K/Newman artifacts, and created the real GitHub Issue. No screenshot was invented because no standalone FR-10 screenshot existed in the Step K evidence set. Test expectations were not changed.

Human Review:

- Human Gate H: APPROVED by the user
- Confirmed defect: **FR10-BUG-01**
- Severity: High
- Related testcases: **AI-FR10-003**, **AI-FR10-021**, **AI-FR10-040**
- GitHub Issue: <https://github.com/nhphunng/SoftwareTesting-Homework/issues/27>
- Step L: COMPLETE

AI Audit Entry - AI-037

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T09:29:00+07:00
Stage	API 3 / FR-16 — Step A Requirement Extraction
API	POST /api/admin/import-products
User Prompt	@Liebe Dùng \$api-testing-human-loop bắt đầu FR16 — Step A: Requirement Extraction cho POST /api/admin/import-products , theo đúng workflow trong plan.md.
Evidence/Input Basis	source/api_specification.md ; /Users/nguyenhoangphihung/Document/eshop-sut/README.md FR-12/FR-15/FR-16 and SEC-01..SEC-07; plan.md ; .agents/skills/api-testing-human-loop/SKILL.md ; .agents/skills/api-testing-human-loop/references/workflow-gates.md ; reports/api-inventory.md ; reports/api-selection.md ; /Users/nguyenhoangphihung/Document/eshop-sut/backend/server.js as implementation observation only

Field	Content
Generated/Modified Files	<code>PoolC-FR-16-ImportProducts/analysis/requirements.md</code> ; <code>reports/ai-session-logs/AI-037.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Extracted FR-16 Step A and stopped at Human Gate A. Preserved the material contract conflict: FR-16 requires CSV file upload with <code>.csv</code> , exact header, RFC4180 parsing, row validation, atomic rollback, and clear result reporting, while <code>api_specification.md</code> documents JSON <code>{products:[...]}</code> for the same endpoint. Kept exact HTTP status codes/response schema and FR-15 inheritance unresolved. Mapped SEC-02 and SEC-03 as directly applicable. Recorded current server behavior only as implementation observations and runtime targets, not confirmed defects. No runtime execution or evidence was fabricated. Full output preserved in <code>reports/ai-session-logs/AI-037.md</code> .

Human Review:

- Status: APPROVED
- Accepted: Gate A approved; dual-surface model retained for FR-16 CSV business contract and JSON `products[]` API contract.
- Modified:
- Removed:
- Added:
- Notes: Proceed to Step B — Domain Partition Design for both input surfaces.

AI Audit Entry - AI-038

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T09:38:00+07:00
Stage	API 3 / FR-16 — Gate A approval + Step B Domain Partition Design
API	<code>POST /api/admin/import-products</code>
User Decision / Prompt	approve Gate A, bước tiếp theo đúng workflow sẽ là Step B — Domain Partition Design cho cả hai input surface: CSV business contract và JSON <code>products[]</code> API contract.
Evidence/Input Basis	Approved <code>PoolC-FR-16-ImportProducts/analysis/requirements.md</code> ; <code>plan.md</code> Step B; <code>.agents/skills/api-testing-human-loop/references/workflow-gates.md</code> ; FR-16/FR-12/SEC requirements already extracted at Step A
Generated/Modified Files	<code>PoolC-FR-16-ImportProducts/analysis/requirements.md</code> ; <code>PoolC-FR-16-ImportProducts/analysis/domain-partitions.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Recorded Human Gate A as approved and created the FR-16 dual-surface domain model. CSV coverage includes file presence/extension, exact header variants, RFC4180 quoted-comma and parser cases, per-row name/price partitions, unclear description/image/category fields, batch composition, and first/middle/last invalid-

Field	Content
	row rollback partitions. JSON coverage includes top-level <code>products[]</code> shape, row type/shape, field-level name/price/description/image/category partitions, malformed/additional inputs, and batch atomicity. Auth partitions explicitly cover SEC-02 and the direct SEC-03 role boundary between admin and valid non-admin JWTs. No undocumented file-size, row-count, name-length, category-range, response-status, or parser limits were promoted to mandatory requirements. Gate B remains pending human review. No runtime execution or evidence was created.

Human Review:

- Status: APPROVED
- Accepted: Gate B approved; dual-surface partition model, rollback row-position partitions, and direct non-admin role boundary retained.
- Modified:
- Removed:
- Added:
- Notes: Proceed to Step C — State Transition / Persistence Analysis.

AI Audit Entry - AI-039

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T09:45:00+07:00
Stage	API 3 / FR-16 — Gate B approval + Step C State Transition / Persistence Analysis
API	<code>POST /api/admin/import-products</code>
User Decision / Prompt	Tôi approve Gate B, bước tiếp theo đúng workflow thực hiện StepC - State Trasition / Persistence Analysis
Evidence/Input Basis	Approved <code>PoolC-FR-16-ImportProducts/analysis/requirements.md</code> ; approved <code>PoolC-FR-16-ImportProducts/analysis/domain-partitions.md</code> ; <code>plan.md</code> Step C; <code>.agents/skills/api-testing-human-loop/references/workflow-gates.md</code>
Generated/Modified Files	<code>PoolC-FR-16-ImportProducts/analysis/domain-partitions.md</code> ; <code>PoolC-FR-16-ImportProducts/analysis/state-transitions.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Recorded Gate B as approved and modeled FR-16 using persistence states/sequences rather than inventing a named lifecycle state machine. Defined baseline, fully persisted, rolled-back, forbidden partial-persistence, unauthorized-unchanged, and repeat-import characterization states. Added valid single/multi-row transitions; invalid first/middle/last-row rollback sequences; auth/role state-preservation sequences; CSV-to-JSON bridge reasoning; repeated import characterization; failed→valid recovery; valid batch A→invalid batch B transactional isolation; parser/body failure persistence rules; optional concurrency characterization; and a before/after runtime verification strategy. Partial persistence is explicitly forbidden by direct FR-16 atomicity.

Field	Content
	Duplicate/idempotency/concurrency/status/schema details remain unresolved. Gate C remains pending human review. No runtime execution or evidence was fabricated.

Human Review:

- Status: APPROVED
- Accepted: Gate C approved; persistence-state model, forbidden partial persistence, before/after persistence oracle, authorization-preserving transitions, and characterization-only duplicate/concurrency treatment retained.
- Modified:
- Removed:
- Added:
- Notes: Proceed to Step D — Security Analysis.

AI Audit Entry - AI-040

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T09:51:00+07:00
Stage	API 3 / FR-16 — Gate C approval + Step D Security Analysis
API	<code>POST /api/admin/import-products</code>
User Decision / Prompt	Approve Gate C; perform Step D — Security Analysis for FR16, focusing on SEC-02, SEC-03, import authorization, injection/persistence risks, and CSV/import-specific attack surface.
Evidence/Input Basis	Approved FR-16 Step A requirements; approved Step B domain partitions; approved Step C persistence/state model; <code>plan.md</code> Step D; <code>\$api-testing-human-loop</code> Gate D guidance; current <code>backend/server.js</code> inspected as implementation observation only
Generated/Modified Files	<code>PoolC-FR-16-ImportProducts/analysis/state-transitions.md</code> ; <code>PoolC-FR-16-ImportProducts/analysis/security.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Recorded Gate C as approved and created FR-16 Security Analysis. Mapped SEC-01..SEC-07 by actual applicability: SEC-02 and SEC-03 directly apply, SEC-05 applies to the persistence path, SEC-04 is only partially verifiable at API scope, and SEC-01/06/07 are not applicable. Added authentication/tampered-token coverage, direct non-admin role-escalation tests, role-claim tampering, persistence/atomicity security cases, SQL-oriented inert payloads with no-unintended-mutation oracle, CSV formula-injection and parser/header/column-smuggling risks, JSON duplicate-key/type-confusion/unexpected-field characterization, information-disclosure characterization, resource-abuse characterization, replay/duplicate-policy boundaries, transport-conflict security reasoning, and before/after persistence security oracles. Generic IDOR/ownership was explicitly excluded because no user-owned path resource exists. Static source observations were kept separate from

Field	Content
	defect confirmation. Gate D remains pending human review. No runtime execution or evidence was fabricated.

Human Review:

- Status: APPROVED
- Accepted: Gate D approved; direct SEC-02/SEC-03 coverage, SEC-05 persistence-path probes, partial SEC-04 treatment, CSV/import risk characterization, and separation of implementation observations from confirmed defects retained.
- Modified:
- Removed:
- Added:
- Notes: Proceed to Step E — Schema / Response Analysis.

AI Audit Entry - AI-041

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T09:55:00+07:00
Stage	API 3 / FR-16 — Gate D approval + Step E Schema / Response Analysis
API	<code>POST /api/admin/import-products</code>
User Decision / Prompt	Approve Gate D; perform Step E — Schema / Response Analysis, explicitly separating the FR-16 business report (<code>success count / error count / reasons</code>) from actual/documented endpoint response behavior without inventing exact schema/status codes.
Evidence/Input Basis	Approved FR-16 Step A-D artifacts; <code>source/api_specification.md</code> §6/§6.3; <code>plan.md</code> Step E; <code>\$api-testing-human-loop</code> schema/response checkpoint guidance; current <code>backend/server.js</code> response shape inspected as implementation observation only
Generated/Modified Files	<code>PoolC-FR-16-ImportProducts/analysis/security.md</code> ; <code>PoolC-FR-16-ImportProducts/analysis/schema.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Recorded Gate D as approved and created FR-16 Schema / Response Analysis. Preserved three layers: FR-16 semantic business-report obligations, API specification request contract with no defined response schema/status, and current implementation response characterization. Treated <code>success count</code> , <code>error count</code> , and failure reasons as semantic requirements without inventing JSON property names. Kept exact success/validation/auth/authorization/parser statuses, response field names/types/nesting, row-number formatting, Content-Type, and error-count interpretation unresolved. Current implementation fields <code>message</code> , <code>inserted</code> , <code>errors</code> , and <code>error</code> remain S2 runtime/implementation characterization only. Added response-counter vs persistence consistency rules, invalid-batch rollback semantics, CSV-vs-JSON response responsibility separation, auth/role response policy, information-disclosure characterization, a three-layer S1/S2/S3 schema strategy, and

Field	Content
	16 candidate schema/response tests. Schema/response checkpoint remains pending human review. No runtime execution or evidence was fabricated.

Human Review:

- Status: APPROVED
- Accepted: Schema/Response checkpoint approved; three-layer S1/S2/S3 response model, semantic report obligations, persistence-first atomicity oracle, and unresolved exact status/schema boundaries retained.
- Modified:
- Removed:
- Added:
- Notes: Proceed to Step F — AI Testcase Generation.

AI Audit Entry - AI-042

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T10:09:00+07:00
Stage	API 3 / FR-16 — Step F AI Testcase Generation
API	<code>POST /api/admin/import-products</code>
User Decision / Prompt	Step F — AI Testcase Generation: create ≥ 35 AI-generated testcases for FR16 based on the approved Requirement + Domain + State + Security + Schema analyses.
Evidence/Input Basis	Approved <code>analysis/requirements.md</code> ; approved <code>analysis/domain-partitions.md</code> ; approved <code>analysis/state-transitions.md</code> ; approved <code>analysis/security.md</code> ; approved <code>analysis/schema.md</code> ; <code>\$api-testing-human-loop</code> testcase/evidence contract
Generated/Modified Files	<code>PoolC-FR-16-ImportProducts/analysis/schema.md</code> ; <code>PoolC-FR-16-ImportProducts/generated/PoolC-FR16-ImportProducts-test.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Recorded the Schema/Response checkpoint as approved and generated 52 AI-origin FR-16 testcase records (requirement ≥ 35). Coverage includes 6 Functional, 15 Domain, 6 Boundary, 8 State/Sequence, 12 Security, and 5 Schema cases. The suite covers canonical JSON imports, CSV business workflow, exact headers/RFC4180 cases, request-shape partitions, name/price boundaries, first/middle/last-row rollback, failure-recovery sequences, SEC-02 authentication, SEC-03 role escalation, token tampering, SEC-05 inert SQL probes, CSV parser/formula risks, duplicate JSON keys/type confusion, and S1/S2 response characterization. Every official-request testcase carries the assignment header requirement. Exact undocumented HTTP statuses and response schemas remain UNRESOLVED. Human audit fields are not filled by AI; execution/evidence/actual-result/defect fields remain blank. No runtime execution or evidence was fabricated.

Human Review:

- Status: COMPLETE
- Accepted: Tester completed Step G human audit for all 52 AI-generated FR-16 cases: 45 VALID, 7 INCOMPLETE, 0 INVALID.
- Modified: Human Audit Status and Human Audit Reason were populated in `PoolC-FR-16-ImportProducts/generated/PoolC-FR16-ImportProducts-test.md` from the tester-provided reviewed artifact.
- Removed:
- Added: INCOMPLETE cases are `AI-FR16-016`, `AI-FR16-017`, `AI-FR16-022`, `AI-FR16-023`, `AI-FR16-030`, `AI-FR16-040`, and `AI-FR16-046`.
- Notes: Corrected Test remains pending for the 7 INCOMPLETE cases. Do not relabel them as VALID until the tester reviews and approves explicit corrections/policy dispositions.

AI Audit Entry - AI-043

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T10:21:00+07:00
Stage	API 3 / FR-16 — Step G Human Audit synchronization
API	<code>POST /api/admin/import-products</code>
User Input	Tester supplied a reviewed <code>PoolC-FR16-ImportProducts-test.md</code> containing per-case Human Audit Status and Human Audit Reason.
Evidence/Input Basis	Tester-provided reviewed artifact with 52 AI cases; existing generated FR-16 testcase artifact; <code>\$api-testing-human-loop</code> provenance rules
Generated/Modified Files	<code>PoolC-FR-16-ImportProducts/generated/PoolC-FR16-ImportProducts-test.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Synchronized the tester's human audit into the project artifact without changing testcase provenance. Final human-audit distribution is 45 VALID / 7 INCOMPLETE / 0 INVALID. Preserved the tester's reasons verbatim for every testcase. The seven INCOMPLETE cases are 016, 017, 022, 023, 030, 040, and 046. Corrected Test fields remain blank; no AI-generated correction is represented as human-approved. Execution/Actual Result/Evidence/Defect fields remain untouched.

Human Review:

- Status: COMPLETE — source is the tester-provided reviewed artifact
- Accepted: 45 VALID / 7 INCOMPLETE / 0 INVALID
- Notes: Next action should resolve the 7 INCOMPLETE cases before treating the complete 52-case suite as implementation-ready.

AI Audit Entry - AI-044

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T10:29:00+07:00

Field	Content
Stage	API 3 / FR-16 — Step G INCOMPLETE-case correction proposal + AI re-review
API	<code>POST /api/admin/import-products</code>
User Prompt	Resolve the 7 INCOMPLETE cases by proposing Corrected Test for each, then re-review whether they can become VALID.
Evidence/Input Basis	Tester-provided Human Audit Status/Reason for all 52 AI cases; approved FR-16 Requirement/Domain/State/Security/Schema analyses; provenance rules from <code>\$api-testing-human-loop</code>
Generated/Modified Files	<code>PoolC-FR-16-ImportProducts/generated/PoolC-FR16-ImportProducts-test.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Added AI-proposed Corrected Test and explicit AI Re-review rows to exactly the seven human-INCOMPLETE cases while preserving their original Human Audit Status/Reason. Cases 016, 017, 022, 023, 030, and 040 are assessed READY FOR HUMAN RE-REVIEW because the corrections add concrete inputs and/or branching persistence-integrity oracles without inventing unresolved acceptance policies. Case 046 remains NOT READY FOR VALID and is safely reclassified as BLOCKED until Step J identifies a concrete, reproducible, non-destructive DB-error trigger. No Human Audit Status was changed by AI. No execution evidence was produced.

Human Review:

- Status: PARTIALLY APPROVED — 6 corrections approved; 1 remains blocked
- Approved by tester: **AI-FR16-016, 017, 022, 023, 030, 040**
- Still blocked: **AI-FR16-046**
- Notes: Tester approved the six proposed corrections; their Human Audit Status is now VALID — HUMAN RE-REVIEW APPROVED. **AI-FR16-046** remains INCOMPLETE/BLOCKED.

AI Audit Entry - AI-045

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T10:41:00+07:00
Stage	API 3 / FR-16 — Human re-review approval of corrected Step G cases
API	<code>POST /api/admin/import-products</code>
User Decision / Prompt	Tester explicitly approved the six proposed corrections for AI-FR16-016, 017, 022, 023, 030, and 040 .
Evidence/Input Basis	Tester approval in chat; AI-044 correction proposals; existing human-reviewed FR-16 testcase artifact
Generated/Modified Files	<code>PoolC-FR-16-ImportProducts/generated/PoolC-FR16-ImportProducts-test.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Updated the six tester-approved corrected cases from INCOMPLETE to VALID — HUMAN RE-REVIEW APPROVED , preserved their AI provenance, retained the

Field	Content
	accepted Corrected Test, and recorded human approval. Suite human-audit distribution is now 51 VALID / 1 INCOMPLETE / 0 INVALID. AI-FR16-046 remains INCOMPLETE/BLOCKED because no safe deterministic DB-error trigger exists yet. No runtime execution/evidence was created.

Human Review:

- Status: APPROVED
- Accepted: **AI-FR16-016**, **AI-FR16-017**, **AI-FR16-022**, **AI-FR16-023**, **AI-FR16-030**, **AI-FR16-040** corrected tests
- Remaining incomplete: **AI-FR16-046** — BLOCKED pending safe deterministic DB-error trigger
- Final distribution at this checkpoint: 51 VALID / 1 INCOMPLETE / 0 INVALID

AI Audit Entry - AI-046

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T10:47:00+07:00
Stage	API 3 / FR-16 — Step H Human-added testcase integration
API	POST /api/admin/import-products
User Decision / Prompt	Integrate the tester's six proposed human-authored cases into the FR-16 testcase artifact after reviewing and correcting their scope/oracles.
Evidence/Input Basis	Tester-authored six-case artifact; approved FR-16 analyses; completed Step G human audit/re-review; prior review finding that HUM-FR16-002 is out of FR-16 scope and the other five need narrowed or branching oracles
Generated/Modified Files	PoolC-FR-16-ImportProducts/generated/PoolC-FR16-ImportProducts-test.md; reports/ai-audit-report.md
AI Output	Added five genuinely human-authored Step H cases with Source = HUMAN : HUM-FR16-001 concurrent duplicate race, HUM-FR16-003 stale category reference characterization, HUM-FR16-004 post-logout same-JWT lifecycle characterization, HUM-FR16-005 stale pre-promotion token after role change, and HUM-FR16-006 repeated invalid-batch submissions. Narrowed their oracles to avoid inventing undocumented uniqueness, referential-integrity, token-revocation, or abuse-control policy. HUM-FR16-002 was intentionally excluded from FR-16 because its primary action/oracle belongs to Product CRUD ownership/authorization. Step H minimum ≥ 5 human-added cases is satisfied. No runtime execution/evidence was created.

Human Review:

- Status: SOURCE CASES AUTHORED BY TESTER; INTEGRATED WITH REVIEWED CORRECTIONS
- Retained: **HUM-FR16-001**, **HUM-FR16-003**, **HUM-FR16-004**, **HUM-FR16-005**, **HUM-FR16-006**
- Excluded from FR-16 scope: **HUM-FR16-002**
- Step H count: 5 human-authored cases
- Notes: Execution Status / Actual Result / Evidence / Defect ID remain blank pending real execution.

AI Audit Entry - AI-047

Field	Content
AI Tool	OpenAI Codex
Date and Time	2026-08-21T11:09:14+07:00
Stage	API 3 / FR-16 — Step I Postman implementation
API	POST /api/admin/import-products
User Decision / Prompt	Implement all executable VALID FR-16 cases in Postman; retain AI-FR16-046 as BLOCKED; preserve IDs/provenance and unresolved response behavior; use persistence oracles; implement five HUMAN records including real concurrency; block missing cross-endpoint dependencies; do not execute Newman or generate evidence.
Evidence/Input Basis	Approved FR-16 requirements, domain, state, security, and schema analyses; 52 AI testcase records with Step G dispositions; five retained Step H HUMAN records; supplied API specification; existing FR-05/FR-10 Postman conventions
Generated/Modified Files	PoolC-FR-16-ImportProducts/postman/build-collection.mjs; PoolC-FR-16-ImportProducts/postman/implementation.md; postman/collection/HW06-FR16-ImportProducts.postman_collection.json; postman/environment/HW06-FR16-Local.postman_environment.json; reports/ai-audit-report.md
AI Output	Implemented 43 executable AI testcase IDs and three executable HUMAN testcase IDs with 56 Postman request items, before/after product persistence oracles, blank credential/category runtime variables, branching characterization checks, sequential state cases, five repeated invalid submissions, and genuine parallel dispatch for HUM-FR16-001 . Preserved all 57 testcase records in the collection, including eight VALID CSV cases technically blocked because no real CSV transport is documented, AI-FR16-046 blocked without a fabricated DB trigger, and HUMAN cases 004/005 blocked because required logout/revocation and role-promotion APIs do not exist in the supplied contract. Applied X-Student-Id: 23127194 to every executable primary/helper request. Performed static JSON/header/ID/script validation only; no Newman/SUT execution or evidence was produced.

Human Review:

- Status: PENDING — Step I artifacts await tester review before Step J/runtime preparation.
- Provenance preserved: 52 **AI-FR16-*** records and five retained **HUM-FR16-*** records.
- Official execution: NOT STARTED.
- Evidence fields: NOT POPULATED.

AI Audit Entry - AI-047

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T11:17:00+07:00

Field	Content
Stage	API 3 / FR-16 — Step J Runtime Test Data / Preconditions
API	<code>POST /api/admin/import-products</code>
User Decision / Prompt	Proceed to Step J — Prepare Runtime Test Data / Preconditions after Step I completion.
Evidence/Input Basis	Step I collection/environment/implementation notes; live EShop backend; live category/user state; existing git-ignored HW06 runtime JWTs; backend route inventory; frontend-admin CSV implementation source
Generated/Modified Files	<code>scripts/prepare-fr16-runtime.py</code> ; <code>postman/environment/HW06-FR16-Local.private.postman_environment.json</code> (git-ignored); <code>postman/data/FR16-runtime-data.json</code> ; <code>PoolC-FR-16-ImportProducts/postman/runtime-fixture-manifest.json</code> ; <code>PoolC-FR-16-ImportProducts/postman/runtime-preconditions.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Verified the live SUT endpoint; re-verified real admin/non-admin JWT roles; generated tampered/forged invalid JWT fixtures without exposing secrets; selected real category ID 1; verified absent category ID 2147483647; created a real disposable category for HUM-FR16-003; wrote a private git-ignored FR16 environment and non-secret runtime metadata. Source inspection confirmed the real CSV workflow exists in frontend-admin via browser FileReader/client-side parsing followed by JSON POST, but no callable CSV upload/multipart API exists, so the 8 CSV cases remain blocked for faithful Newman execution. Backend route inventory also confirms no logout/revocation route and no role-promotion route, leaving HUM-FR16-004/005 blocked; AI-FR16-046 remains blocked for lack of a safe deterministic DB-error trigger. 46 testcase IDs are runtime-ready. No official Newman run or testcase verdict/evidence was produced.

Human Review:

- Status: PENDING STEP J REVIEW
- Runtime-ready: 43 AI + 3 HUMAN testcase IDs
- Blocked: 8 CSV AI cases + AI-FR16-046 + HUM-FR16-004 + HUM-FR16-005
- Notes: Proceed to Step K only after reviewing this Step J preparation.

AI Audit Entry - AI-048

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T11:29:00+07:00
Stage	API 3 / FR-16 — Step K Official Newman Execution + Gate G preparation
API	<code>POST /api/admin/import-products</code>
User Decision / Prompt	Tester approved Step J and requested Step K using <code>\$api-testing-human-loop</code> and the Step K runner.

Field	Content
Evidence/Input Basis	Approved Step J runtime setup; <code>scripts/run-step-k.sh</code> ; real live Newman run; raw <code>FR16-official-*</code> evidence; compact execution summary; approved FR-16 contract/security/atomicity oracles
Generated/Modified Files	<code>postman/newman/FR16-official-cli.txt</code> ; <code>postman/newman/FR16-official-report.json</code> ; <code>postman/newman/FR16-official-report.html</code> ; <code>postman/newman/FR16-official-report.xml</code> ; <code>postman/newman/FR16-execution-summary.json</code> ; <code>postman/newman/FR16-execution-summary.md</code> ; <code>scripts/extract-newman-summary.js</code> ; <code>PoolC-FR-16-ImportProducts/evidence/step-k-execution.md</code> ; <code>reports/ai-audit-report.md</code>
AI Output	Performed the official FR16 Newman run. Final raw execution: 46 testcase IDs, 195 requests, 388 assertions, 342 passed assertions, 46 failed assertions, 17 testcase IDs with failures, and exact <code>X-Student-Id: 23127194</code> on 195/195 Newman-observed requests. Newman exit code was 1. A compact-summary reporting defect was discovered because the extractor recognized <code>HUMAN-FR..</code> but not FR16's <code>HUM-FR..</code> ; the extractor was corrected and the summary was regenerated from the same raw Newman JSON without rerunning the SUT. Gate G preparation groups runtime failures into potential FR16 validation/atomic-rollback defects and a potential FR12/SEC-03 admin-role authorization defect. The 11 documented Step J blockers were not executed and are not treated as failures. No bug was confirmed or filed; Human Gate H remains pending.

Human Review:

- Status: PENDING GATE G / GATE H REVIEW
- Real execution evidence: COMPLETE
- Potential defect cluster A: FR-16 invalid-row validation / atomic rollback
- Potential defect cluster B: non-admin actor can import through admin endpoint
- Notes: Do not create bug reports or GitHub Issues until the tester confirms the defect grouping and Gate H conclusions.

AI Audit Entry - AI-049

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T11:43:00+07:00
Stage	API 3 / FR-16 — Human Gate G approval and defect-candidate grouping
API	<code>POST /api/admin/import-products</code>
User Decision / Prompt	Tester approved the Step K execution evidence, approved the <code>HUM-FR</code> extractor fix, accepted all 11 blocked cases as NOT EXECUTED, selected Option B, and confirmed three potential product defects as Gate H candidates.
Evidence/Input Basis	Raw FR16 Newman evidence; corrected compact summary; approved FR-16/FR-12 requirements; Step K Gate G classification

Field	Content
Generated/Modified Files	PoolC-FR-16-ImportProducts/evidence/step-k-execution.md ; reports/ai-audit-report.md
AI Output	Recorded Gate G as human-approved. Preserved three distinct defect candidates for Human Gate H: (1) FR16 accepts non-positive prices, (2) FR16 violates atomic all-or-nothing rollback, and (3) non-admin users can import through the admin endpoint. The reporting-tool issue is accepted as fixed. The 11 blocked testcase IDs remain NOT EXECUTED and are neither PASS nor FAIL. No bug report or GitHub Issue has been created yet.

Human Review:

- Status: GATE G APPROVED
- Candidate 1: FR16 accepts non-positive prices
- Candidate 2: FR16 violates atomic all-or-nothing rollback
- Candidate 3: non-admin user can import through [/api/admin/import-products](#)
- Blocked cases: 11 accepted as NOT EXECUTED
- Next gate: Human Gate H — confirm or reject each candidate as a genuine defect

AI Audit Entry - AI-050

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T11:55:00+07:00
Stage	API 3 / FR-16 — Human Gate H approval + Step L confirmed defect reporting
API	POST /api/admin/import-products
User Decision / Prompt	Tester approved all three Gate H defect conclusions and requested Step L bug reporting.
Evidence/Input Basis	Human-approved Gate G Option B grouping; approved FR-16/FR-12 requirements; real FR16 Newman evidence; Step K persistence results; corrected compact summary
Generated/Modified Files	PoolC-FR-16-ImportProducts/bugs/FR16-BUG-01.md ; FR16-BUG-02.md ; FR16-BUG-03.md ; reports/ai-audit-report.md ; GitHub Issues #28, #29, #30
AI Output	Confirmed and documented three genuine defects after Human Gate H approval: (1) non-positive prices are accepted/persisted, (2) invalid import batches violate atomic all-or-nothing rollback, and (3) a valid non-admin JWT can import through the admin endpoint. Created three Markdown bug reports and GitHub Issues #28/#29/#30. Screenshot workflow was documented but no screenshot was fabricated or claimed to exist.

Human Review:

- Status: GATE H APPROVED / STEP L BUG REPORTING COMPLETE
- Confirmed defect: [FR16-BUG-01](#) — GitHub #28
- Confirmed defect: [FR16-BUG-02](#) — GitHub #29

- Confirmed defect: **FR16-BUG-03** — GitHub #30
- Screenshot evidence: PENDING HUMAN CAPTURE FROM REAL NEWMAN/POSTMAN EVIDENCE VIEW

AI Audit Entry - AI-051

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T20:40:00+07:00
Stage	API 3 / FR-16 — Step L screenshot evidence capture
API	POST /api/admin/import-products
User Decision / Prompt	Tester requested Newman evidence screenshots for the three confirmed FR16 defects.
Evidence/Input Basis	Real postman/newman/FR16-official-report.json ; confirmed Gate H defect mapping; representative testcases AI-FR16-019 , AI-FR16-024 , AI-FR16-035
Generated/Modified Files	PoolC-FR-16-ImportProducts/evidence/screenshots/FR16-BUG-01.png ; FR16-BUG-02.png ; FR16-BUG-03.png ; matching *-evidence.html files; three FR16 bug Markdown reports; reports/ai-audit-report.md
AI Output	Rendered three evidence views directly from the real official Newman JSON and captured them as PNG using local headless Chrome. Each image identifies the testcase, approved oracle, observed persistence result, failed Newman assertions, and X-Student-Id: 23127194 . Authorization/JWT values are intentionally redacted. The views explicitly identify the raw Newman JSON as the source of truth and are not represented as native Postman UI screenshots.

Human Review:

- Status: SCREENSHOT EVIDENCE CREATED FROM REAL NEWMAN RUN
- FR16-BUG-01 representative evidence: **AI-FR16-019**
- FR16-BUG-02 representative evidence: **AI-FR16-024**
- FR16-BUG-03 representative evidence: **AI-FR16-035**
- JWT exposure: REDACTED

AI Audit Entry - AI-052

Field	Content
AI Tool	ChatGPT with artifact_tool spreadsheet generator + Liebe local coding connector
Date and Time	2026-08-21T21:08:00+07:00
Stage	HW06 — P9 Consolidate Excel Test Suite
API	Consolidated FR05 + FR10 + FR16
User Decision / Prompt	Tester requested P9 — consolidate the completed API testcase suites into a single Excel workbook.
Evidence/Input Basis	Authoritative testcase Markdown for FR05/FR10/FR16; FR10 human-authored testcase file; real Newman execution summaries for all three APIs; Human Gate H

Field	Content
	confirmed bug reports.
Generated/Modified Files	Generated HW06-API-Test-Cases.xlsx as the P9 consolidated workbook artifact; updated reports/ai-audit-report.md .
AI Output	Consolidated 152 testcase records across all three selected APIs while preserving provenance and execution semantics: 136 AI-generated and 16 HUMAN-authored; 119 PASS, 22 FAIL, 11 FR16 cases explicitly NOT EXECUTED — BLOCKED ; seven confirmed defect records mapped to supporting testcase IDs. Workbook contains Summary , All Test Cases , FR05 , FR10 , FR16 , Defects , and Sources sheets, including execution-status conditional formatting, authoritative Markdown source paths, evidence references, and a summary chart. Formula/error validation completed with no spreadsheet formula errors. The Excel workbook is an export/index artifact; the Markdown testcase files remain the source of truth for full testcase definitions.

Human Review:

- Status: P9 EXCEL CONSOLIDATION GENERATED — PENDING TESTER REVIEW
- Total testcase records: 152
- AI-generated: 136
- HUMAN-authored: 16
- PASS: 119
- FAIL: 22
- NOT EXECUTED — BLOCKED: 11
- Confirmed defects: 7
- Source-of-truth rule: detailed testcase definitions remain in their authoritative Markdown artifacts

AI Audit Entry - AI-053

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T21:11:00+07:00
Stage	HW06 — P10 Finalize Postman/Newman
API	Consolidated FR05 + FR10 + FR16
User Decision / Prompt	Tester requested P10 — finalize the Postman/Newman artifacts after P9 consolidation.
Evidence/Input Basis	Three Postman collections; three public environments; git-ignored private runtime environments; three runtime-data files; real Step K Newman CLI/JSON/HTML/JUnit reports and compact summaries; shared run-step-k.sh runner.
Generated/Modified Files	postman/README.md ; reports/postman-newman-finalization.md ; scripts/validate-postman-newman.mjs ; reports/ai-audit-report.md
AI Output	Finalized the Postman/Newman package without rerunning the SUT. Verified all required collection/environment/data/Newman evidence files for FR05, FR10, and FR16; confirmed public collections/environments contain no JWT-like literals;

Field	Content
	confirmed private runtime environments are git-ignored; confirmed each collection has a collection-level pre-request script enforcing exact X-Student-Id: 23127194 ; and confirmed the real Newman evidence records the exact student header on 353/353 observed requests. Preserved truthful official outcomes: 805 assertions total, 749 passed, 56 failed. FR16's 11 blocked testcase records remain NOT EXECUTED — BLOCKED , neither PASS nor FAIL. Added a reusable static/evidence validation script and documented Postman/Newman features and runner usage.

Human Review:

- Status: P10 FINALIZED — PENDING TESTER REVIEW
- Collections: 3/3 present
- Public environments: 3/3 present, no JWT literals found
- Private runtime environments: 3/3 git-ignored
- Runtime data files: 3/3 present
- Newman HTML/JSON/CLI/JUnit evidence: 3/3 feature sets present
- Student header evidence: 353/353 exact match
- Validation: **P10_VALIDATION=PASS**
- New SUT execution during P10: NONE
- Next recommended phase: P11 — CI/CD integration

AI Audit Entry - AI-054

Field	Content
AI Tool	ChatGPT with Liebe local coding connector + GitHub Actions / GitHub CLI
Date and Time	2026-08-21T21:30:00+07:00
Stage	HW06 — P11 CI/CD Integration
API	CI demonstration covering FR05 + FR10 + FR16
User Decision / Prompt	Tester requested P11 — integrate Newman into CI/CD and produce the required all-pass and exactly-one-fail pipeline samples.
Evidence/Input Basis	Phase 6 requirements in plan.md ; finalized Postman/Newman artifacts from P10; public EShop SUT repository; real GitHub Actions runs and downloaded artifacts/logs.
Generated/Modified Files	.github/workflows/hw06-api-tests.yml ; cicd/postman/HW06-CI-Demo.postman_collection.json ; pass/one-fail environments ; cicd/verify-ci-demo.mjs ; cicd/ci-mode.txt ; cicd/CI-CD-Report.md ; cicd/evidence/run-a/** ; cicd/evidence/run-b/** ; cicd/screenshots/run-a-all-pass.png ; cicd/screenshots/run-b-one-fail.png ; reports/ai-audit-report.md
AI Output	Implemented a GitHub Actions Newman pipeline on branch hw06 using a controlled CI demonstration suite that is explicitly separate from official Step K defect evidence. Run A commit 06cd7d9 produced real GitHub Actions run 32492084678 with SUCCESS, 4 requests, 12 assertions, 0 failed, and exact X-Student-Id: 23127194

Field	Content
	on 4/4 requests. Run B commit b564ce2 produced real run 32492204368 ; the verifier confirmed exactly one failure (CI-DEMO-ONLY) among 13 assertions with SID 4/4, then the workflow intentionally marked the pipeline red and stated that the red run is not a new SUT defect. Commit 582811c restored mode to pass and run 32492332236 completed SUCCESS. Real GitHub logs/Newman artifacts were downloaded and GitHub Actions run-page screenshots were captured.

Human Review:

- Status: P11 COMPLETE — PENDING TESTER REVIEW
- Run A all-pass: SUCCESS — <https://github.com/nhphunng/SoftwareTesting-Homework/actions/runs/32492084678>
- Run A Newman: 12 assertions / 0 failed / SID 4/4
- Run B intentional one-fail: FAILURE — <https://github.com/nhphunng/SoftwareTesting-Homework/actions/runs/32492204368>
- Run B Newman: 13 assertions / exactly 1 failed / SID 4/4
- Intentional failure label: **CI-DEMO-ONLY** | **intentional single failure**
- Restore run: SUCCESS — <https://github.com/nhphunng/SoftwareTesting-Homework/actions/runs/32492332236>
- Official Step K evidence modified: NO
- New SUT bug claimed from Run B: NO
- Next recommended phase: P12 — AI-driven API Test Generator

AI Audit Entry - AI-055

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T21:40:00+07:00
Stage	HW06 — P12 AI-Driven API Test Generator
API	Reusable generator; demo on POST /api/admin/import-products / FR16; regression on FR05 and FR10
User Decision / Prompt	Tester requested the next planned step, P12 — implement the AI-Driven API Test Generator / Agent Skill.
Evidence/Input Basis	plan.md Phase 7 requirements; source/api_specification.md ; explicit demo security input SEC-02 valid JWT required ; SEC-03 admin role required ; existing human-in-the-loop provenance rules.
Generated/Modified Files	agent-skill/README.md ; agent-skill/pseudocode.md ; agent-skill/api-test-generator/SKILL.md ; agent-skill/api-test-generator/README.md ; agent-skill/api-test-generator/generate.mjs ; agent-skill/demo/fr16-generator-output.json ; agent-skill/demo/fr16-generator-output.md ; reports/ai-audit-report.md
AI Output	Implemented a dependency-free deterministic Node.js generator that accepts a spec path, selected endpoint/FR, explicit security requirements, and JSON/Markdown output format. It produces a requirement model, endpoint model, coverage model, AI

Field	Content
	candidate tests, source hash/audit metadata, explicit unresolved assumptions, and <code>finalizationGate = HUMAN_REVIEW_REQUIRED</code> . The FR16 demo generated 8 candidates and deliberately did not invent positive-price validation or atomic rollback because those rules are absent from <code>source/api_specification.md</code> . Regression also generated candidates for FR05 and FR10.

Human Review / Correction:

- Initial demo exposed a generator defect: the response-status regex misclassified the `100` prefix of example `price: 10000` as HTTP status 100.
- Correction applied: HTTP status extraction now requires a recognized HTTP reason phrase such as `200 OK`, preventing ordinary numeric business values from becoming status codes.
- Duplicate body-field modeling for the top-level `products` array was also corrected.
- Post-correction FR16 result: documented HTTP statuses = none; exact response schema = unresolved; 8 candidates; 7 unresolved candidates; all candidates source = `AI_CANDIDATE`; gate = `HUMAN_REVIEW_REQUIRED`.
- Anti-hallucination check: no `price > 0`, positive-price, all-or-nothing, or atomic-rollback rule was generated from the API specification alone.
- Regression: FR05 candidate generation PASS; FR10 candidate generation PASS; FR16 candidate generation PASS.
- Runtime evidence fabricated/generated by the skill: NO.
- Final self-drawn diagram generated by AI: NO — intentionally deferred because the assignment requires student-drawn final evidence.
- Status: P12 IMPLEMENTATION COMPLETE — PENDING TESTER REVIEW.
- Next planned step: P13 — self-drawn diagram + diagram/pseudocode finalization.

AI Audit Entry - AI-056

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T22:25:00+07:00
Stage	HW06 — P13 Self-Drawn Diagram + Pseudocode Finalization
API	AI-Driven API Test Generator architecture
User Decision / Prompt	Tester confirmed the Mermaid diagram had been manually reviewed/edited and requested closing P13 by moving the final diagram out of <code>drafts/</code> , rerendering it, updating the audit, and committing/pushing.
Evidence/Input Basis	Student-reviewed Mermaid source <code>agent-skill/drafts/api-test-generator-diagram.mmd</code> ; previously finalized <code>agent-skill/pseudocode.md</code> ; P12 generator architecture and Human Review Gate design.
Generated/Modified Files	Finalized <code>agent-skill/api-test-generator-diagram.mmd</code> ; finalized <code>agent-skill/api-test-generator-diagram.png</code> ; removed temporary <code>agent-skill/drafts/</code> ; updated <code>reports/ai-audit-report.md</code> .
AI Output	Promoted the student-reviewed Mermaid source and PNG from the draft area into the final P13 artifact paths. Re-rendered the final <code>.mmd</code> with Mermaid CLI and verified

Field	Content
	the output PNG SHA-256 was byte-identical to the student's exported PNG. Confirmed the diagram contains the required architecture stages: Requirement Parser, Endpoint Model, Parameter/Constraint Analyzer, Domain Partition, State Transition, Security Mapper, Schema Generator, Coverage Analyzer, Deduplicator, AI Candidates, Human Review Gate, Final Structured API Test Cases, and Audit Metadata. Cross-checked the same logical stages against <code>pseudocode.md</code> .

Human Review / Finalization:

- Student manually reviewed/edited the Mermaid diagram before finalization: YES.
- Final Mermaid source: `agent-skill/api-test-generator-diagram.mmd`.
- Final PNG: `agent-skill/api-test-generator-diagram.png`.
- Final PNG reproducible from final Mermaid source: YES.
- Render comparison: SHA-256 identical.
- Pseudocode consistency review: PASS.
- AI-generated final diagram claim: NO — AI supplied the earlier draft/blueprint; the final artifact was student-reviewed/edited before promotion.
- Runtime evidence fabricated/generated: NO.
- Status: P13 COMPLETE.
- Next planned step: P14 — Compile AI Audit Report.

AI Audit Entry - AI-057

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T22:30:00+07:00
Stage	HW06 — P14 Compile AI Audit Report
API	Cross-assignment audit compilation for FR05 + FR10 + FR16 + P9–P13
User Decision / Prompt	Tester requested <code>P14 — Compile AI Audit Report</code> .
Evidence/Input Basis	Existing chronological <code>reports/ai-audit-report.md</code> ; existing <code>reports/ai-session-logs/AI-037.md</code> ; existing <code>reports/ai-session-logs/FR10-step-c-summary.md</code> ; Phase 8 requirements in <code>plan.md</code> ; previously generated execution/CI/generator/diagram artifacts referenced by the historical audit.
Generated/Modified Files	<code>reports/ai-audit-report.md</code> ; <code>reports/ai-session-logs/README.md</code> ; <code>scripts/validate-ai-audit.mjs</code>
AI Output	Compiled the existing audit without reconstructing missing history. Added a P14 compilation summary, required-field normalization map, corpus coverage counts, evidence-integrity rules, Human-AI collaboration chains, and an anomaly register for duplicate historical IDs <code>AI-010</code> and <code>AI-047</code> . Indexed only the two session-log artifacts that actually exist and added a reusable validator. Historical entry text and numbering were preserved to avoid breaking traceability. PDF export remains intentionally deferred because the plan says to export it later.

Human Review / Correction:

- Historical interaction occurrences before P14: 59.
- Distinct historical IDs before P14: 56.
- Historical numeric range **AI-001..AI-056**: no missing numeric ID.
- Duplicate historical IDs preserved transparently: **AI-010** x3 and **AI-047** x2.
- Required Tool/Date-Time/Prompt/Output/Artifact/Human Review presence was validated across the historical corpus before P14.
- Missing transcripts reconstructed: NO.
- Historical user decisions fabricated: NO.
- Runtime evidence changed: NO.
- Confirmed defect evidence changed: NO.
- Status: P14 COMPILED — PENDING TESTER REVIEW.
- Next planned step: P15 — Evidence-Based AI Critique.

AI Audit Entry - AI-058

Field	Content
AI Tool	ChatGPT with Liebe local coding connector
Date and Time	2026-08-21T22:33:00+07:00
Stage	HW06 — P15 Evidence-Based AI Critique
API	Cross-assignment critique covering FR05 + FR10 + FR16
User Decision / Prompt	Tester requested P15 — Write Evidence-Based AI Critique (200–300 words) .
Evidence/Input Basis	Phase 9 requirements in plan.md ; FR05 audit summary and human-added testcase reasons; FR10 audit/human-added cases and confirmed shipping-state defect; FR16 human audit, blocked case, human-added cases, and confirmed price/rollback/role-authorization defects; P14 audit compilation.
Generated/Modified Files	reports/AI-Critique.md ; reports/ai-audit-report.md
AI Output	Wrote a 250-word critique grounded in actual HW06 evidence. It distinguishes AI incompleteness from invalidity, cites the 10 FR05 / 6 FR10 / 7 FR16 initially INCOMPLETE cases, identifies cross-endpoint/compositional reasoning as a major blind spot, cites HUMAN-FR05-044 and HUMAN-FR05-048 as human-added cases that exposed three FR05 defects, acknowledges additional human coverage dimensions, and explicitly notes that AI did detect the FR10 shipping-state defect and the major FR16 validation/rollback/authorization defects once deterministic runtime fixtures were available.

Human Review / Correction:

- Required length: 200–300 words.
- Actual critique length: 250 words, excluding the Markdown heading.
- Generic praise-only critique: NO.
- Actual INVALID evidence: no AI-generated testcase ultimately remained INVALID; critique states this explicitly instead of inventing an INVALID example.

- INCOMPLETE evidence included: YES — FR05=10, FR10=6, FR16=7 before correction/re-review.
- Human-added evidence included: YES.
- Bugs AI missed included: YES — FR05 defects triggered by HUMAN-FR05-044 and HUMAN-FR05-048.
- Bugs AI did catch acknowledged: YES — FR10 shipping-state cancellation and FR16 price/rollback/role-authorization defects.
- Prompt limitation discussed: YES — predominantly single-endpoint, partition-driven generation reduced temporal/parser-layer/cross-feature exploration.
- Status: P15 COMPLETE — PENDING TESTER REVIEW.
- Next planned step: P16 — Main Report + README.